

TACHS Exam Simulator

Full-Length Practice Exam 01

Complete printable test with 200 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Reading	50	40 minutes
Written Expression	50	40 minutes
Mathematics	50	45 minutes
Ability	50	35 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 160 minutes. Calculators should be used only when permitted by the rules followed for your TACHS preparation.

Section 1: Reading

50 questions | Recommended time: 40 minutes

Literary Passage: The Unfinished Sign

Mara had promised the neighborhood garden committee that its new wooden sign would be ready by Saturday. On Friday afternoon, however, the painted letters looked stiff and uneven. She had copied them carefully from a computer printout, yet the sign seemed more suited to an office hallway than to a garden filled with curling vines and bright zinnias.

Her grandfather found her sanding away a blot of green paint. "The letters are readable," he said.

"Readable is not the same as welcoming," Mara replied.

Instead of starting over, she walked through the garden. She noticed how bean tendrils looped around their poles and how sunflower leaves leaned at different angles. Back in the shed, she softened the straight edges of the letters and added small painted leaves between the words. One leaf tilted too far, and a drop of yellow landed beneath it. Mara nearly wiped it away. Then she turned the drop into a tiny bee.

The next morning, visitors paused at the gate to admire the sign. A child pointed first to the bee, not the words. Mara smiled. The mistake she had almost erased had given the sign the lively quality she had been seeking.

1. Why does Mara walk through the garden?

- A) To observe details that could improve the sign
- B) To ask visitors what the sign should say
- C) To find a different board for the sign
- D) To decide where the committee should plant flowers

2. What can most reasonably be inferred about Mara?

- A) She prefers computer work to painting.
- B) She cares about how her work makes people feel.
- C) She expects her grandfather to finish difficult projects.
- D) She is reluctant to change a plan once she begins.

3. Which statement best expresses a central idea of the passage?

- A) Careful copying is the best way to make attractive artwork.
- B) Public projects should always be completed ahead of schedule.
- C) Observation and flexibility can turn an imperfection into a strength.
- D) Natural objects are easier to paint than letters.

4. As used in the first paragraph, "stiff" most nearly means

- A) difficult to bend physically.
- B) sore from working too long.
- C) thick because the paint had dried.
- D) formal and lacking liveliness.

Informational Passage: A Library for Useful Objects

At a traditional library, members borrow books. At a “library of things,” they may borrow a drill, a telescope, a sewing machine, or a set of camping pans. These lending programs have appeared in many communities because people often need certain objects only briefly. Buying each item can be expensive, and storing rarely used equipment takes space.

The system resembles ordinary library lending. Members reserve an item, collect it for a stated period, and return it so another person can use it. Some programs charge a yearly fee, while others are supported by donations or local funding. Before borrowing specialized equipment, members may have to attend a short safety lesson.

The benefits extend beyond saving money. Sharing objects can reduce the number of products manufactured and later discarded. The programs may also encourage neighbors to exchange knowledge. A person borrowing a sewing machine, for example, might attend a repair workshop taught by an experienced volunteer.

Running such a library requires careful planning. Tools must be inspected, missing parts replaced, and popular objects scheduled fairly. Yet supporters argue that these challenges are manageable. To them, a shelf holding one well-used tool is more valuable than many closets each holding an identical tool that is almost never touched.

5. According to the passage, why might a borrower be required to attend a lesson?

- A) To learn how to use specialized equipment safely
- B) To qualify for a lower yearly membership fee
- C) To help inspect every object in the collection
- D) To teach another member a repair skill

6. The author includes the sewing-machine example mainly to show that libraries of things can

- A) replace traditional libraries in small communities.
- B) encourage people to share skills as well as objects.
- C) operate successfully without staff members.
- D) lend household equipment more easily than tools.

7. How is the passage mainly organized?

- A) It presents events in the order in which they occurred.
- B) It compares two competing businesses and selects a winner.
- C) It explains a concept, describes its benefits, and acknowledges its challenges.
- D) It states a problem and gives detailed instructions for solving it.

8. As used in the final paragraph, “manageable” most nearly means

- A) profitable.
- B) temporary.
- C) avoidable.
- D) possible to handle successfully.

Humanities or Reflective Passage: The Value of a Slow Look

In a museum, visitors sometimes move from painting to painting as though they are collecting proof that they saw each one. A quick glance may reveal a subject, a dominant color, or a familiar name. It rarely reveals how the parts of a work guide the eye or how a small detail changes the mood.

Looking slowly does not require expert knowledge. A viewer can begin with simple questions: Where did my eyes land first? Which line or color led them elsewhere? What detail did I miss at first? These questions shift attention from identifying the picture to experiencing how it works.

Slow looking can also make uncertainty useful. A puzzling expression on a painted face may invite several explanations. Rather than rushing to read the museum label, a viewer can gather evidence from posture, setting, and light. The label may later provide context, but it need not replace observation.

This habit matters beyond museums. Careful attention helps people notice assumptions in advertisements, patterns in scientific observations, and changes in familiar places. Speed is valuable when a decision must be made quickly. Yet when understanding matters more than coverage, giving one object sustained attention can reveal more than hurrying past twenty.

9. Which statement best summarizes the passage?

- A) Sustained observation can deepen understanding in museums and everyday life.
- B) Museum labels prevent viewers from forming independent opinions.
- C) People need expert training to interpret unfamiliar paintings.
- D) Quick decisions are usually less accurate than slow decisions.

10. The author would most likely agree that museum labels should

- A) be removed from paintings that contain puzzling details.
- B) provide context after viewers have first made their own observations.
- C) explain the single correct emotional response to each work.
- D) be read before viewers look at a painting.

11. What is the author's main purpose in the passage?

- A) To criticize museums for displaying too many works
- B) To compare artistic observation with scientific measurement
- C) To encourage readers to practice deliberate, evidence-based attention
- D) To prove that paintings are more valuable than advertisements

12. Which question does the passage specifically recommend that a viewer ask?

- A) How much did the artist earn for this work?
- B) Why did the museum purchase this painting?
- C) Which other visitors noticed the same feature?
- D) Where did my eyes land first?

13. Why does the author mention advertisements and scientific observations?

- A) To show that careful attention is useful in settings beyond art
- B) To suggest that both should be displayed in museums
- C) To argue that visual information is always misleading
- D) To identify subjects that require faster decisions

Literary Passage: The Last Rehearsal

For weeks, Jonah had practiced the opening melody for the school orchestra concert. During the final rehearsal, however, his clarinet produced a thin squeak on the third note. He tried again, with the same result. Around him, chairs scraped as the other musicians settled into place.

Ms. Alvarez examined the instrument and discovered a torn pad beneath one key. There was no time to replace it before rehearsal began. "You can use the spare clarinet," she said, handing Jonah an older instrument from the cabinet.

The spare felt unfamiliar. Its keys resisted his fingers, and its tone seemed darker than the one he knew. Jonah wanted to ask someone else to play the opening. Instead, he took a slow breath and tested the difficult measures quietly. By the third attempt, his hands had adjusted.

When Ms. Alvarez raised her baton, Jonah began. The first phrase was not as polished as it had sounded at home, but it was steady. Later, during a rest, he noticed that Priya, the new violinist, was staring anxiously at her music. Jonah caught her eye and nodded toward the conductor. Priya straightened and lifted her bow.

After rehearsal, Jonah returned the spare clarinet. He had not played perfectly, yet the instrument no longer felt like an obstacle. It had become proof that preparation could include adapting when a plan changed.

14. What problem does Ms. Alvarez discover?

- A) A pad beneath one of Jonah's clarinet keys is torn.
- B) Jonah has practiced the wrong opening melody.
- C) The orchestra cabinet has no spare instruments.
- D) Priya's violin needs a new string.

15. Why does Jonah nod toward the conductor when he sees Priya looking anxious?

- A) He wants Priya to ask the conductor for help.
- B) He is encouraging Priya to focus and continue.
- C) He is warning Priya that rehearsal is almost over.
- D) He wants Priya to notice a mistake in the music.

16. Which statement best expresses a central idea of the passage?

- A) Older instruments usually produce richer sounds than newer ones.
- B) Musicians perform best when they practice without an audience.
- C) Responding calmly to an unexpected problem can strengthen confidence.
- D) Helping another person is more important than preparing for a concert.

17. As used in the third paragraph, "resisted" most nearly means

- A) made a loud sound.
- B) appeared damaged.
- C) moved too quickly.
- D) required extra effort to press.

Scientific Passage: How Fog Collectors Gather Water

In some dry coastal regions, rain is scarce even though thick fog frequently moves across hillsides. Fog collectors can capture some of the water carried in that air. A typical collector consists of a fine mesh panel held upright by a frame. As wind pushes fog through the mesh, tiny droplets strike the threads and join together. Once the drops become heavy enough, gravity pulls them into a channel below the panel. From there, the water flows into a storage container.

The amount collected depends on several factors. A site needs regular fog and enough wind to move moist air through the mesh. The threads must be spaced closely enough to catch droplets but not so closely that they block airflow. Researchers also test different materials and coatings to help drops slide downward rather than remain on the panel.

Fog collectors do not create water, and they cannot supply every community with all the water it needs. Their usefulness varies by location and season. Still, they can provide water for gardens, tree seedlings, or other limited purposes where suitable conditions exist. Because the panels require no fuel to operate, they may serve as a simple addition to other water sources. Careful site testing remains essential: a collector placed where fog is rare may gather almost nothing, no matter how well the mesh is designed.

18. What causes collected droplets to move from the mesh into the channel?

- A) Heat
- B) Gravity
- C) A fuel-powered pump
- D) A protective coating

19. Why does the author state that fog collectors “do not create water”?

- A) To explain why the mesh must be replaced often
- B) To compare the collectors with rain gauges
- C) To argue that collectors should be used only for research
- D) To clarify a limitation and prevent an exaggerated impression of the technology

20. How is the passage mainly organized?

- A) It explains how a device works, identifies factors affecting it, and discusses its practical limits.
- B) It lists several inventions in the order they were developed.
- C) It contrasts two experiments and explains why one failed.
- D) It gives step-by-step directions for building a collector at home.

21. As used in the final paragraph, “suitable” most nearly means

- A) inexpensive.
- B) permanent.
- C) appropriate for the purpose.
- D) easy to reach.

Informational Historical-Style Passage: Messages Across the Mountain

Before electronic communication reached many mountain communities, sending an urgent message could require a rider to travel for hours. In one region, a group of towns created a chain of signal stations on high ridges. The system described here is a historical-style example rather than a record of one specific network.

Each station could see the station before it and the one after it. Operators used pairs of movable wooden arms mounted on a tall post. Different arm positions represented letters, numbers, or short instructions. An operator viewed an incoming signal through a telescope, recorded it, and then repeated it for the next station. In clear weather, a message could cross the region far faster than a horse could carry it.

Speed did not guarantee accuracy. A tired operator might copy an angle incorrectly, and mist could hide a station entirely. To reduce mistakes, operators paused after a group of symbols and waited for a confirmation signal. Important messages were also written in a log so that the sequence could be checked later.

The network required trained workers and unobstructed views, which made it expensive to maintain. Eventually, telegraph wires offered a more dependable way to send messages in poor visibility and after dark. Even so, the ridge stations demonstrated an important principle: information can travel quickly when a long journey is divided among connected points.

22. According to the passage, what did an operator do after viewing an incoming signal?

- A) Rode to the next station with a written copy
- B) Changed the meaning into a shorter instruction
- C) Waited until darkness before forwarding it
- D) Recorded the signal and repeated it for the next station

23. The author's tone toward the signal system is best described as

- A) amused by its unnecessary complexity.
- B) appreciative of its ingenuity while aware of its weaknesses.
- C) doubtful that such a system could transmit any information.
- D) critical of the towns for replacing riders.

24. Which statement best summarizes the passage?

- A) A chain of visual stations sped communication by relaying coded signals, although weather and human error limited the system.
- B) Mountain towns used telescopes mainly to predict dangerous weather for travelers.
- C) Telegraph wires were developed because horses could not cross mountain roads.
- D) Operators preferred written logs because arm signals represented only numbers.

25. Why was the confirmation signal especially useful?

- A) It showed operators how to repair the wooden arms.
- B) It allowed stations to work after dark.
- C) It helped detect an incorrectly received group of symbols before the message continued.
- D) It warned riders that a message was urgent.

Literary Passage: The Second Launch

Mara had tested her model boat three times in the bathtub, but the park pond looked far less forgiving. A breeze wrinkled the water, and reeds leaned over the launch spot. She set the cedar boat down. It traveled barely a yard before turning in a slow circle.

"The sail is crooked," her cousin Eli said.

Mara wanted to blame the wind. Instead, she pulled the boat ashore and examined the mast. The sail was straight, but the small metal weight beneath the hull had shifted to one side. She remembered fastening it hurriedly after dinner.

Using a coin as a screwdriver, Mara loosened the weight and centered it. The second launch began badly too: a gust pushed the boat toward the reeds. Then the sail caught steady air, and the boat moved across the pond in a clean diagonal line.

Eli ran along the bank, cheering. Mara stayed near the launch spot, watching the wake narrow behind the boat.

She had imagined success as proof that her design was perfect. Instead, the best moment had come from discovering exactly where it was not. She opened her notebook and wrote, "Test outdoors earlier." Below that she added, "Bring a real screwdriver."

26. What caused the boat to turn in a circle during its first launch?

- A) The weight beneath the hull had shifted.
- B) The sail had torn near the mast.
- C) Reeds were caught around the rudder.
- D) Eli had placed the boat facing the wrong way.

27. What does Mara's decision to examine the boat instead of blaming the wind show?

- A) She doubts that outdoor testing is useful.
- B) She is willing to investigate her own mistake.
- C) She prefers Eli's judgment to her own.
- D) She expects the boat to fail completely.

28. Why does the author include Mara's two notebook entries at the end?

- A) To suggest that Mara plans to build a larger boat
- B) To show that Eli will lead the next experiment
- C) To emphasize the practical lessons Mara gained
- D) To explain why the pond was a poor testing site

29. In the passage, "the pond looked far less forgiving" most nearly means that the pond

- A) seemed darker than the bathtub
- B) was difficult to reach safely
- C) appeared unusually shallow
- D) would expose weaknesses more readily

Literary Passage: Before the Doors Opened

On Saturday morning, Luis stood inside the neighborhood museum holding a stack of blank labels. His job was to place them beside objects in a new exhibit about everyday inventions. The curator had arranged an early alarm clock, a folding ruler, and a hand-cranked eggbeater in glass cases.

Luis began copying descriptions from a folder. At the eggbeater, he paused. The description explained its gears but not why anyone had valued it. He pictured visitors glancing at the object and moving on.

In a storage box, Luis found a note donated with the eggbeater. Its owner recalled using it every Sunday while her brother slowly poured batter into a bowl. Luis asked the curator whether he could add one sentence from the memory. She read the note, then nodded.

When the museum doors opened, a family stopped at the case. A boy turned the handle of an identical eggbeater mounted outside the glass. His grandfather pointed to the label and described making pancakes without an electric mixer. Luis watched them talk until the next visitors arrived.

He had thought labels simply named and explained things. Now he understood that a few careful words could also open a door between an object and someone's own experience.

30. What did Luis find in the storage box?

- A) A replacement handle for the eggbeater
- B) A note from the eggbeater's former owner
- C) A photograph of the first museum exhibit
- D) A list of objects the curator had rejected

31. Why does Luis pause after reading the original eggbeater description?

- A) He believes the description lacks a human connection.
- B) He notices that the object is in the wrong case.
- C) He cannot understand how the gears operate.
- D) He worries that the label contains a spelling error.

32. Which statement best expresses the central idea of the passage?

- A) Old tools are usually harder to explain than modern devices.
- B) Museum visitors learn best when they can touch every object.
- C) Brief personal details can make historical objects meaningful.
- D) Curators depend on volunteers to complete exhibits on time.

33. How does the family's conversation near the end contribute to the passage?

- A) It introduces a disagreement that Luis must resolve.
- B) It proves that the curator's original description was inaccurate.
- C) It shifts attention away from Luis to the museum's founder.
- D) It demonstrates the effect of the sentence Luis added.

Informational Passage: Why Cities Plant Tiny Forests

Some cities are creating dense patches of native trees on plots no larger than a tennis court. These “tiny forests” differ from ordinary parks. Instead of spacing a few mature trees across open lawns, planners place many young trees and shrubs close together. The plants compete for sunlight, encouraging rapid upward growth, while fallen leaves remain on the ground and enrich the soil.

Before planting, teams study nearby natural woodlands to identify native species. They may loosen compacted soil and mix in compost so roots can spread. Volunteers then plant several layers of vegetation, including low shrubs, medium-height trees, and species that may eventually form a canopy.

Tiny forests cannot replace large natural forests. Their small size limits the wildlife they can support, and young plantings require watering and weeding. However, they can cool a neighborhood, absorb rainwater, provide habitat for insects and birds, and give residents a nearby place to observe seasonal change.

Researchers continue to measure how well these forests perform in different climates. The most useful comparisons follow planted sites for years and compare them with similar unplanted areas. Such studies can help cities decide where dense planting offers the greatest benefit and where another use of limited land would serve residents better.

34. According to the passage, why are young trees planted close together?

- A) Competition for light encourages upward growth.
- B) Close planting eliminates the need for watering.
- C) Dense roots prevent leaves from reaching the soil.
- D) Crowded trees remain shorter than park trees.

35. Why should researchers compare planted sites with similar unplanted areas?

- A) To determine which native trees grow tallest
- B) To separate the forest's effects from other local conditions
- C) To calculate how many volunteers visited each location
- D) To prove that every unused plot should become a forest

36. Which description best characterizes the author's tone?

- A) Doubtful and dismissive
- B) Amused and informal
- C) Favorable but carefully qualified
- D) Urgent and alarmed

37. Which statement best summarizes the passage?

- A) Tiny forests are dense urban plantings with potential benefits that require careful planning and long-term study.
- B) Tiny forests grow without maintenance because native plants always thrive in cities.
- C) Cities should replace ordinary parks with compact forests whenever land is available.
- D) Researchers have already proven that tiny forests support the same wildlife as large forests.

38. As used in the passage, “compacted” most nearly means

- A) enriched with nutrients
- B) shaded by plants
- C) divided into layers
- D) pressed tightly together

Literary Passage: The Missing Beat

Nia had practiced the drum pattern until her hands knew it without help. Yet during rehearsal, the dancers repeatedly reached the stage's center one beat after her final strike. Mr. Chen asked everyone to begin again. Nia played louder, certain that a stronger rhythm would solve the problem. The dancers still arrived late. During the break, Nia sat at the edge of the stage and tapped the pattern against her knee. Across the room, Mateo silently practiced the dancers' opening steps. Nia noticed that he paused before the third turn to avoid a loose strip of tape on the floor. Each dancer made the same tiny adjustment. Nia peeled up the tape and asked Mateo to try the steps again. Without the obstacle, he reached center exactly on her final tap. When rehearsal resumed, the whole group landed together. "Good catch," Mr. Chen said. Nia looked at her drum. She had assumed that any timing problem belonged to the musician. Playing louder had only made the mismatch easier to hear. The solution had come when she watched what the rhythm was guiding.

39. Why did Mateo pause before the third turn?

- A) He was listening for a louder drumbeat.
- B) He was waiting for the other dancers.
- C) He was avoiding a loose strip of tape.
- D) He had forgotten the next step.

40. What does Nia learn from solving the rehearsal problem?

- A) She should observe the entire situation before deciding what caused a problem.
- B) Musicians should change their tempo whenever dancers make a mistake.
- C) Small stage problems are always more important than careful practice.
- D) A louder performance makes it easier for a group to work together.

41. Which statement best summarizes the passage?

- A) A drummer improves a difficult pattern by practicing during a break.
- B) A dancer teaches a musician how to repair part of a stage.
- C) A rehearsal succeeds because the group begins playing more loudly.
- D) Nia discovers that an apparent timing error is caused by an obstacle affecting the dancers.

42. In the last sentence, the word "guiding" most nearly means

- A) correcting
- B) directing
- C) following
- D) measuring

Scientific Passage: Soil's Hidden Waterways

After a heavy rain, water does not move through every soil at the same speed. The size of the spaces between soil particles helps determine what happens. Sandy soil contains relatively large spaces. Water can pass through them quickly, so sand drains well but may not hold moisture near plant roots for long. Clay soil has much smaller spaces. It often absorbs water slowly and can remain wet after rain.

Organic matter, such as decomposed leaves, changes this picture. Mixed into sandy soil, it acts somewhat like a sponge, holding water that might otherwise drain away. In dense clay, organic matter helps create larger pathways through which water and air can travel.

Earthworms also influence these underground routes. As they tunnel, they form channels that allow water to enter soil rather than flowing across its surface. Their castings add nutrients and contribute organic material.

Gardeners cannot easily replace all the soil in a yard, but they can improve how it handles water. Adding compost, avoiding repeated walking on damp ground, and keeping living roots in the soil can protect the spaces water needs. Healthy soil is not merely a pile of particles; it is a changing network of passages.

43. According to the passage, why does sandy soil drain quickly?

- A) Its particles contain large spaces between them.
- B) Its organic matter blocks water near the surface.
- C) Its earthworms remove moisture from plant roots.
- D) Its particles absorb more water than clay does.

44. What is the author's main purpose in the passage?

- A) To argue that gardeners should replace clay with sand
- B) To compare the nutrients required by different plants
- C) To explain how soil structure affects water movement and how that structure can be improved
- D) To describe an experiment measuring the number of earthworms in a garden

45. How is the passage mainly organized?

- A) It presents a gardening problem and tells a story about one successful gardener.
- B) It describes factors affecting water movement, then gives ways to protect or improve soil structure.
- C) It lists soil types in order from least useful to most useful.
- D) It compares two experiments and explains why their results conflict.

46. As used in the final paragraph, “dense” most nearly means

- A) deep
- B) fertile
- C) dry
- D) tightly packed

Biographical-Style Passage: Measuring a Moving Shore

When coastal surveyor Lena Ortiz began her career, measuring a beach meant carrying poles, notebooks, and a heavy leveling instrument across the sand. Her team returned to the same marked locations each season. By comparing the height and position of the shoreline over time, they could tell where storms had removed sand and where currents had deposited it.

Years later, Ortiz began using small camera-equipped aircraft to record the coast from above. Software combined hundreds of images into detailed surface models. The new method covered more ground and revealed patterns that isolated measurements could miss. It did not, however, make fieldwork unnecessary. Plants could hide the sand, shadows could confuse the software, and waves could erase a temporary marker. Ortiz still walked the beach to check unusual results.

She also invited local students to place simple measuring stakes beyond the reach of ordinary tides. Their monthly photographs created a steady record between the team's larger surveys. Ortiz valued the project not only for its data but also for the questions students began asking about dunes, storms, and building near the shore.

For Ortiz, better instruments did more than increase speed. They made it possible to connect precise measurements, direct observation, and community attention to a coastline that never remained still.

47. Why does Ortiz still walk the beach after adopting aerial surveys?

- A) To train the aircraft to follow the shoreline
- B) To move student stakes closer to the water
- C) To check unusual results that images or software may produce
- D) To replace surface models with handwritten maps

48. What can reasonably be inferred about Ortiz's view of technology?

- A) It is most useful when combined with observation and human judgment.
- B) It should replace older methods as soon as it becomes available.
- C) It produces reliable results only when students operate it.
- D) It matters more than public understanding of the coast.

49. The author's tone toward Ortiz's work is best described as

- A) doubtful and impatient
- B) respectful and thoughtful
- C) humorous and informal
- D) alarmed and critical

50. Which statement best expresses the central idea of the passage?

- A) Coastal surveys became less accurate when aircraft replaced leveling instruments.
- B) Student photographs provide all the information researchers need about beaches.
- C) Shorelines can be protected only by measuring them after major storms.
- D) Ortiz combines evolving tools, field checks, and community participation to study coastal change.

Section 2: Written Expression

50 questions | Recommended time: 40 minutes

Revision Passage: A Cooler Schoolyard

[1] On sunny afternoons, the paved area beside our school becomes uncomfortably hot. [2] Our environmental club proposes adding three shade trees along its western edge. [3] Mature trees would block some afternoon sunlight and provide a more comfortable place for students to gather. [4] The trees could also absorb rainwater that now flows rapidly toward the storm drain. [5] Last year, the drama club performed a comedy in the auditorium. [6] Native tree species should be selected because they are suited to local conditions. [7] Club members can help water the young trees, but a grounds worker should supervise their long-term care. [8] With careful planning, this small project could make the schoolyard cooler and manage rainwater more effectively.

51. Which sentence should be removed because it does not support the passage's central idea?

- A) Sentence 3
- B) Sentence 5
- C) Sentence 6
- D) Sentence 7

52. Which revision of sentence 2 most clearly identifies what "its" refers to?

- A) Our environmental club proposes adding three shade trees along the paved area's western edge.
- B) Along its western edge, three shade trees are proposed by our environmental club.
- C) Our environmental club, along its western edge, proposes three additional shade trees.
- D) Three shade trees are what our environmental club proposes adding along its western edge.

53. Which transition would best begin sentence 4?

- A) In contrast,
- B) For instance,
- C) In addition,
- D) Nevertheless,

54. Which revision best combines sentences 6 and 7 while preserving their meaning?

- A) Native species are suited to local conditions, club members water them and a grounds worker supervises care.
- B) Because native species, suited to local conditions, and club members can help with watering.
- C) Club members can water native species suited to local conditions, a grounds worker supervising long-term care.
- D) Because native species suit local conditions, club members can help water them, while a grounds worker supervises long-term care.

Revision Passage: Recording Family Recipes

[1] My aunt cooks several family dishes without consulting written recipes. [2] She judges quantities by sight and changes seasonings after tasting. [3] I decided to record her methods so younger relatives could prepare the same dishes. [4] At first, I simply wrote “add enough water,” which would not help someone unfamiliar with the meal. [5] During our next cooking session, I measured each ingredient before my aunt added it. [6] I also noted how the sauce looked at different stages. [7] The finished guide includes exact amounts, sensory clues, and my aunt's suggestions for adjusting the flavor. [8] Recording the recipes required patience, it preserved knowledge that might otherwise have been lost.

55. Which revision of “changes seasonings after tasting” in sentence 2 is most precise?

- A) does things to the seasonings after tasting.
- B) adjusts the seasonings according to the flavor.
- C) makes the seasonings different after she tastes them.
- D) fixes what is in the pot after tasting.

56. Where should the following sentence be placed?

“These visual descriptions captured information that measurements alone could not provide.”

- A) Before sentence 1
- B) After sentence 3
- C) After sentence 6
- D) After sentence 8

57. Which revision best corrects sentence 8?

- A) Recording the recipes required patience, but it preserved knowledge that might otherwise have been lost.
- B) Recording the recipes required patience, preserving knowledge, it might otherwise have been lost.
- C) Requiring patience to record the recipes and preserving knowledge that otherwise might be lost.
- D) The recipes required patience to record, it preserved knowledge that otherwise might have been lost.

58. Which sentence would provide the most effective conclusion to the passage?

- A) Some cookbooks contain photographs beside every recipe.
- B) My aunt owns several measuring cups in different sizes.
- C) Cooking is an activity that many families enjoy together.
- D) By translating experience into clear directions, I created a guide that future cooks can actually use.

59. Which sentence is written correctly?

- A) Neither the coaches nor the captain were available for the interview.
- B) Neither the coaches nor the captain was available for the interview.
- C) Neither the coaches or the captain was available for the interview.
- D) Neither the coaches nor the captain have been available for the interview.

60. Which sentence uses parallel structure correctly?

- A) The workshop teaches students to plan a budget, comparing prices, and how to save.
- B) The workshop teaches students budgeting, to compare prices, and savings.
- C) The workshop teaches students to plan a budget, compare prices, and save money.
- D) The workshop teaches students planning a budget, price comparisons, and to save money.

61. Which sentence uses the pronoun correctly?

- A) The science teacher asked Maya and me to organize the samples.
- B) The science teacher asked Maya and I to organize the samples.
- C) The science teacher asked me and Maya's to organize the samples.
- D) The science teacher asked Maya and myself to organize the samples.

62. Which numbered line contains an error in standard written US English?

Line 1: The hikers checked the map before leaving camp.

Line 2: Each of them carried a full water bottle.

Line 3: The narrow trails were clearly marked.

Line 4: By noon, the group have reached the overlook.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

63. Which numbered line contains an error in standard written US English?

Line 1: Our class visited the city archive on Tuesday.

Line 2: The archivist showed we several handwritten letters.

Line 3: We examined the oldest document more carefully than the others.

Line 4: Later, we discussed why such records should be protected.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

Revision Passage: Mapping Quiet Places

[1] Our student council wanted to identify calm places where students could read or work during lunch. [2] First, volunteers measured noise levels in six areas on three different days. [3] They also counted available seats and noted whether each location was sheltered from bad weather. [4] The library corner was consistently quiet, but it had only eight seats. [5] By contrast, the covered courtyard offered thirty seats and was only slightly louder. [6] Several council members enjoy mystery novels. [7] After reviewing the results, the council recommended adding two small tables to the library corner and posting quiet-zone signs in one section of the courtyard. [8] These changes would create more useful choices without requiring the school to build a new room.

64. Which sentence should be deleted because it does not support the passage's central idea?

- A) Sentence 6
- B) Sentence 3
- C) Sentence 5
- D) Sentence 8

65. Which revision of sentence 2 is most precise?

- A) First, volunteers did some tests in a number of places.
- B) First, volunteers used sound meters to record noise in six areas on three separate days.
- C) First, six areas were things that volunteers looked at repeatedly.
- D) First, volunteers carefully handled an interesting school concern.

66. Which revision of sentence 7 uses verb tense correctly?

- A) After reviewing the results, the council recommends adding two small tables and posted quiet-zone signs.
- B) After reviewing the results, the council had recommend adding two small tables and posts quiet-zone signs.
- C) After reviewing the results, the council recommended adding two small tables and posting quiet-zone signs in one section of the courtyard.
- D) After reviewing the results, the council recommending two small tables and had posted quiet-zone signs.

67. Which revision most effectively combines sentences 4 and 5?

- A) The library corner was consistently quiet, it had only eight seats, and the covered courtyard had thirty.
- B) The library corner was quiet with eight seats, the courtyard was covered and only slightly louder with thirty.
- C) Having only eight seats in the quiet library corner and thirty in the courtyard, which was covered.
- D) Although the library corner was consistently quieter, the covered courtyard offered thirty seats instead of only eight.

Revision Passage: Restoring a Stream Bank

[1] Heavy rain had washed soil from the bank of a stream near our community center. [2] A local conservation group invited students to help stabilize the damaged area. [3] Before planting, the leader explained that native grasses have dense roots that hold soil in place. [4] We carried trays of seedlings down the path and arranged them in marked rows. [5] The work was muddy, and everyone stayed focused until each plant was secure. [6] Next spring, the group will return to measure how much of the bank remains covered. [7] This follow-up matters because restoration is a process, not a single afternoon of work. [8] The project showed us that careful observation after planting is as important as the planting itself.

68. Which opening sentence would best introduce the passage?

- A) Outdoor projects can involve many different tools and supplies.
- B) Streams are bodies of moving water found in many regions.
- C) One rainy Saturday, our class learned how planting and later monitoring can protect an eroded stream bank.
- D) The community center offers several activities for students.

69. Which revision of sentence 5 best shows the relationship between its ideas?

- A) Although the work was muddy, everyone stayed focused until each plant was secure.
- B) The work was muddy, or everyone stayed focused until each plant was secure.
- C) The work was muddy, everyone, however, stayed focused until each plant was secure.
- D) The work was muddy, everyone stayed focused until each plant was secure.

70. Which revision of sentence 6 is most concise without changing its meaning?

- A) Next spring, the group will return back again in order to measure the coverage that remains on the bank.
- B) Next spring, the group will return to measure the bank's remaining plant cover.
- C) In the springtime that comes next, the group will be returning for the purpose of measurement.
- D) The group, next spring, will go back to the bank, and the amount covered will be measured by them.

71. Which sentence would provide the most relevant support after sentence 3?

- A) The selected grasses also tolerate the stream bank's usual moisture and sunlight.
- B) The leader had worked with several volunteer groups before ours.
- C) Native plants can have green, brown, or yellow leaves.
- D) We wore old shoes because the path was muddy.

72. Which sentence is written correctly?

- A) Neither the captain nor the players was prepared for the schedule change.
- B) Neither the players nor the captain were prepared for the schedule change.
- C) Neither the captain nor the players were prepared for the schedule change.
- D) Neither the players or the captain was prepared for the schedule change.

73. Which sentence uses pronouns correctly?

- A) The science teacher asked Maya and I to organize the samples.
- B) The science teacher asked Maya and myself to organize the samples.
- C) The science teacher asked I and Maya to organize the samples.
- D) The science teacher asked Maya and me to organize the samples.

74. Which numbered line contains an error in standard written US English?

Line 1: By the time the doors opened, the volunteers had arranged the chairs.

Line 2: Each row was checked to ensure that the aisles remained clear.

Line 3: The coordinator then give every volunteer a final assignment.

Line 4: At noon, the first visitors entered the hall.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

75. Which numbered line contains an error in standard written US English?

Line 1: The two proposals address the same problem.

Line 2: Of the two plans, Marisol's is the most practical.

Line 3: Her plan requires fewer materials and less time.

Line 4: The committee will discuss both options tomorrow.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

76. Which numbered line contains an error in standard written US English?

Line 1: Luis placed the glass bowl carefully on the counter.

Line 2: He checked the recipe before adding the flour.

Line 3: The batter became smooth after several minutes of mixing.

Line 4: He poured it slow into the prepared pan.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

Revision Passage: A Student Repair Station

[1] Our student council proposes creating a small repair station beside the school office. [2] Students could borrow simple tools to tighten a loose binder ring, replace a backpack buckle, or mend a torn notebook cover. [3] The station would be supervised by an adult, and short instruction cards would explain how to use each tool safely. [4] Many damaged supplies are thrown away even though a quick repair would make them useful again. [5] Last year, the cafeteria began serving a new kind of soup on Fridays. [6] A repair station would reduce waste while teaching students to care for the belongings they already own.

77. Which sentence should be removed because it does not support the passage's central idea?

- A) Sentence 2
- B) Sentence 3
- C) Sentence 5
- D) Sentence 6

78. Which revision most effectively combines sentences 2 and 3?

- A) Students could borrow simple tools for small repairs, with adult supervision and instruction cards explaining safe use.
- B) Students could borrow simple tools, they could make repairs, an adult supervising and cards.
- C) Supervised by an adult, the simple tools could borrow students for repairs and safe instructions.
- D) Students could borrow simple tools for repairs and instruction cards, being supervised safely.

79. Which revision of “a quick repair would make them useful again” is most precise?

- A) doing something would improve them
- B) a brief fix could extend their useful life
- C) they could become sort of better afterward
- D) fixing stuff might make a difference

80. Which additional sentence would best support the proposal?

- A) The school office walls were painted during summer vacation.
- B) Some students prefer zippered backpacks to drawstring bags.
- C) Stores sell tools in many sizes and colors.
- D) A monthly record could show how many items were repaired instead of discarded.

Revision Passage: Listening for Frogs

[1] Each spring, volunteers visit local ponds after sunset to listen for frog calls. [2] Different species produce distinct sounds, so trained listeners can identify frogs without seeing them. [3] Volunteers record the place, time, weather, and species they hear. [4] Because the same ponds are checked repeatedly, scientists can compare observations from different years. [5] The information does not answer every question about frog populations. [6] It does, however, help researchers notice changes that may deserve closer study.

81. Which word would most precisely replace “place” in sentence 3?

- A) location
- B) thing
- C) area-like spot
- D) somewhere

82. Which revision best combines sentences 5 and 6?

- A) The information does not answer every question, it helps researchers notice changes.
- B) Although the information does not answer every question about frog populations, it helps researchers notice changes that may deserve closer study.
- C) Not answering every question about frog populations, changes may deserve closer study by researchers.
- D) The information not answering every question but helping researchers notice changes deserving closer study.

83. Which revision of sentence 4 is most concise while preserving its meaning?

- A) Since volunteers check the same ponds again and again repeatedly, comparisons can be made over years.
- B) The same ponds are checked, and years are different for scientists comparing observations.
- C) Repeated checks of the same ponds allow scientists to compare observations across years.
- D) Scientists, comparing observations, have ponds that are repeatedly the same ones.

84. Which sentence is written correctly?

- A) Neither the coaches nor the captain were prepared to cancel practice.
- B) Neither the coaches nor the captain are prepared to cancel practice.
- C) Neither the coaches nor the captain have been prepared to cancel practice.
- D) Neither the coaches nor the captain was prepared to cancel practice.

85. Choose the correct word to complete the sentence: The science teacher gave Maya and _____ permission to use the lab after school.

- A) I
- B) me
- C) myself
- D) mine

86. Which sentence uses parallel structure correctly?

- A) The workshop taught us to measure carefully, to cut safely, and to assemble patiently.
- B) The workshop taught us careful measurement, cutting safely, and to assemble patiently.
- C) The workshop taught us measuring carefully, how to cut safely, and patient assembly.
- D) The workshop taught us measurement, to cut safely, and patient assembly.

87. Which line contains an error in standard written US English?

Line 1: The two hikers checked their map before leaving.

Line 2: Each of them carried a full bottle of water.

Line 3: Along the trail, they stopped to photograph wildflowers.

Line 4: By noon, neither hiker have reached the planned resting place.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

88. Which line contains an error in standard written US English?

Line 1: Our class visited the print shop on Tuesday.

Line 2: The manager demonstrated how the large press works.

Line 3: Afterward, we watched an employee stacked the finished pages.

Line 4: Everyone thanked the staff before returning to school.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

Revision Passage: Designing a Hallway Exhibit

[1] Our history club decided to create an exhibit about the school building. [2] We first searched yearbooks and interviewed former students. [3] These sources revealed that the auditorium once served as a neighborhood meeting hall. [4] We found an old basketball uniform in a cabinet. [5] To help visitors follow the changes over time, we arranged the materials by decade. [6] Each section included photographs, a short caption, and one recorded memory. [7] Because the hallway is narrow, we tested the display before attaching anything to the walls. [8] This trial showed us where people could stop without blocking the doors. [9] The completed exhibit turned an ordinary hallway into a timeline students could explore between classes.

89. Which sentence should be removed because it does not contribute to the passage's central idea?

- A) Sentence 2
- B) Sentence 4
- C) Sentence 6
- D) Sentence 8

90. Which revision most effectively combines sentences 7 and 8?

- A) Because the hallway is narrow, we tested the display before attaching anything to the walls, revealing where people could stop without blocking the doors.
- B) The hallway is narrow, testing the display before attaching anything showed us, where people could stop without blocking the doors.
- C) We tested the display because the hallway is narrow, and before attaching anything to the walls and where people could stop.
- D) Before attaching anything, the narrow hallway was tested by the display, which showed people stopping without blocking doors.

91. Which revision of sentence 6 is most precise and concise?

- A) Each section presented photographs, a brief caption, and one recorded memory.
- B) In every one of the sections, there were photographs along with a short caption and also one recorded memory.
- C) Each and every section had some various photographs, a caption that was short, and a memory that was recorded.
- D) Photographs were in each section, and there was a caption plus a memory there too.

92. Which sentence would best follow sentence 3 to develop its idea?

- A) Some yearbooks had green covers, while others had blue covers.
- B) The club held its meetings every other Wednesday after school.
- C) We used pencils when taking notes so that we could make corrections.
- D) A graduate recalled attending public lectures there before the school added a separate community room.

Revision Passage: Practicing Trail Courtesy

[1] Sharing a trail safely requires attention to other users. [2] Walkers should stay to the right when the path is wide enough. [3] Before passing someone, a cyclist should slow down and give a clear verbal signal. [4] This warning should be spoken early enough to avoid surprising the walker. [5] Dogs can enjoy public trails, and owners should follow posted leash rules. [6] They should also move aside when stopping, especially near curves or narrow bridges. [7] These habits are simple. [8] They make a busy trail more predictable and comfortable for everyone.

93. Which revision of sentence 4 uses the most appropriate formal tone?

- A) The cyclist should call out before getting way too close and startling somebody.
- B) This warning should be given soon enough to prevent the walker from being startled.
- C) The cyclist has to say something early so the walker does not freak out.
- D) Calling out at the last second is not cool because the walker might jump.

94. Which revision most effectively combines sentences 7 and 8?

- A) These habits are simple, a busy trail becomes more predictable and comfortable for everyone.
- B) Being simple habits and making a busy trail more predictable and comfortable for everyone.
- C) These simple habits make a busy trail more predictable and comfortable for everyone.
- D) A busy trail, these habits are simple, is more predictable and comfortable for everyone.

95. Which revision of sentence 6 makes the pronoun reference clearest?

- A) Dog owners should also move aside when stopping, especially near curves or narrow bridges.
- B) They should also move it aside when stopping, especially near curves or narrow bridges.
- C) These should also move aside when it stops, especially near curves or narrow bridges.
- D) Owners should also move them aside when stopping it, especially near curves or narrow bridges.

96. Which sentence is written correctly?

- A) Of the two routes, the forest path is the most shorter.
- B) Of the two routes, the forest path is the shortest one of both.
- C) Of the two routes, the forest path is more shorter.
- D) Of the two routes, the forest path is shorter.

97. Which sentence uses the correct pronoun case?

- A) The science teacher gave Elena and I permission to use the lab.
- B) The science teacher gave Elena and me permission to use the lab.
- C) The science teacher gave she and I permission to use the lab.
- D) The science teacher gave her and myself permission to use the lab.

98. Which line contains an error in standard written US English?

Line 1: The collection of maps was donated last spring.

Line 2: Each map shows a different region of the state.

Line 3: A group of volunteers are preparing labels for them.

Line 4: The display will open to visitors next month.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

99. Which line contains an error in standard written US English?

Line 1: Priya had already packed the supplies.

Line 2: She checked the list before she left home.

Line 3: At the workshop, she discovers that one brush was missing.

Line 4: Her partner offered an extra brush.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

100. Which line contains an error in standard written US English?

Line 1: The kitten slipped quietly through the open gate.

Line 2: Its bright collar was easy to recognize.

Line 3: We followed it across the empty yard.

Line 4: Soon, the kitten returned back to its owner.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

Section 3: Mathematics

50 questions | Recommended time: 45 minutes

101. A tank is $\frac{3}{5}$ full. After 18 liters are added, it is $\frac{9}{10}$ full. What is the tank's total capacity?

- A) 60 liters
- B) 45 liters
- C) 36 liters
- D) 30 liters

102. What is the value of $24 - 3 \times (5 + 1) \div 2$?

- A) 3
- B) 15
- C) 21
- D) 63

103. A jacket's price is reduced by 20%, and the sale price is then reduced by an additional 10%. The final price is what percent of the original price?

- A) 70%
- B) 80%
- C) 72%
- D) 88%

104. The rule for a sequence is “multiply the previous term by 2, then subtract 3.” If the first term is 5, what is the fourth term?

- A) 11
- B) 15
- C) 17
- D) 19

105. Which expression represents “seven less than three times a number n ”?

- A) $3n - 7$
- B) $7 - 3n$
- C) $3(n - 7)$
- D) $7n - 3$

106. A bicycle rental costs \$6 plus \$4 for each hour. If the total charge is \$30, how many hours was the bicycle rented?

- A) 4 hours
- B) 6 hours
- C) 7 hours
- D) 9 hours

- 107. Five quiz scores have a mean of 84. Four of the scores are 78, 81, 85, and 91. What is the fifth score?**
- A) 82
 - B) 83
 - C) 85
 - D) 89
- 108. A bag contains 3 red tiles, 2 blue tiles, and 1 green tile. Two tiles are drawn without replacement. What is the probability that both are red?**
- A) $\frac{1}{6}$
 - B) $\frac{1}{4}$
 - C) $\frac{1}{3}$
 - D) $\frac{1}{5}$
- 109. Two angles of a triangle measure 47° and 68° . What is the measure of the third angle?**
- A) 65°
 - B) 75°
 - C) 85°
 - D) 115°
- 110. A rectangle has a perimeter of 46 centimeters and a length of 15 centimeters. What is its area?**
- A) 105 cm^2
 - B) 120 cm^2
 - C) 225 cm^2
 - D) 240 cm^2
- 111. A 2.4-meter ribbon is cut into pieces that are each 30 centimeters long. How many complete pieces are made?**
- A) 6
 - B) 7
 - C) 8
 - D) 80
- 112. A movie begins at 6:47 p.m. and lasts 1 hour 58 minutes. At what time does it end?**
- A) 8:35 p.m.
 - B) 8:40 p.m.
 - C) 8:44 p.m.
 - D) 8:45 p.m.
- 113. A recipe uses $2\frac{1}{4}$ cups of oats for 18 bars. At the same rate, how many cups of oats are needed for 30 bars?**
- A) $2\frac{3}{4}$ cups
 - B) 3 cups
 - C) $3\frac{1}{4}$ cups
 - D) $3\frac{3}{4}$ cups

114. What is the greatest common factor of 84 and 126?

- A) 42
- B) 21
- C) 14
- D) 7

115. At sunrise, the temperature was -6°C . By noon, it had risen 11°C , and by evening, it had fallen 8°C . What was the evening temperature?

- A) 13°C
- B) -3°C
- C) 3°C
- D) -13°C

116. For the function $y = 3x - 4$, which ordered pair is a solution?

- A) (1, 1)
- B) (2, 4)
- C) (3, 5)
- D) (4, 7)

117. Solve $5(x - 2) = 3x + 8$.

- A) $x = 7$
- B) $x = 8$
- C) $x = 10$
- D) $x = 9$

118. The numbers of minutes four students read on Monday were 18, 24, 31, and 37. On Tuesday, each student read exactly 5 more minutes than on Monday. How does the Tuesday data set compare with the Monday data set?

- A) The mean increases by 5, and the range increases by 5.
- B) The mean increases by 5, and the range stays the same.
- C) The median stays the same, and the range increases by 5.
- D) Both the mean and the range stay the same.

119. A spinner has eight equal sections numbered 1 through 8. What is the probability of spinning a number that is both even and greater than 3?

- A) $\frac{3}{8}$
- B) $\frac{1}{4}$
- C) $\frac{1}{2}$
- D) $\frac{5}{8}$

120. A club recorded the number of books collected during four weeks.

Week 1: 36

Week 2: 48

Week 3: 42

Week 4: 54

What percent of the total books were collected during Week 4?

A) 20%

B) 25%

C) 27%

D) 30%

121. A right triangle has legs of 9 centimeters and 12 centimeters. What is the length of its hypotenuse?

A) 18 cm

B) 21 cm

C) 15 cm

D) 144 cm

122. A circle has a diameter of 14 inches. Using $\frac{22}{7}$ for π , what is the circumference?

A) 44 inches

B) 22 inches

C) 88 inches

D) 154 inches

123. A square has an area of 81 square units. A second square has side lengths twice as long. What is the area of the second square?

A) 162 square units

B) 324 square units

C) 243 square units

D) 648 square units

124. A map scale states that 1 inch represents 18 miles. Two towns are $3\frac{1}{2}$ inches apart on the map. What is the actual distance between them?

A) 54 miles

B) 72 miles

C) 63 miles

D) 81 miles

125. A rectangular storage box is 10 inches long, 6 inches wide, and 4 inches high. How many 1-inch cubes can completely fill the box?

A) 20

B) 120

C) 200

D) 240

126. What is the value of $\frac{3}{5} \div \frac{9}{10}$?

A) $\frac{2}{3}$

B) $\frac{3}{2}$

C) $\frac{6}{5}$

D) $\frac{27}{50}$

127. A store reduces a \$48 item by 25% and then adds \$3 for delivery. What is the final cost?

- A) \$33
- B) \$36
- C) \$39
- D) \$45

128. Which number has exactly three positive factors?

- A) 8
- B) 9
- C) 12
- D) 15

129. If $4(x - 3) = 2x + 10$, what is x ?

- A) 8
- B) 9
- C) 10
- D) 11

130. The rule for a sequence is “multiply the position number by 3, then subtract 2.” What is the 12th term?

- A) 34
- B) 32
- C) 36
- D) 38

131. A taxi ride costs \$4 plus \$2.50 per mile. Which inequality represents the miles m that can be traveled for no more than \$24?

- A) $4m + 2.50 \leq 24$
- B) $2.50m - 4 \leq 24$
- C) $4 + 2.50m \geq 24$
- D) $4 + 2.50m \leq 24$

132. The numbers of books read by five students are 3, 5, 5, 7, and n . If the mean is 6, what is n ?

- A) 8
- B) 12
- C) 14
- D) 10

133. A bag contains 3 red, 2 blue, and 1 green token. Two tokens are drawn without replacement. What is the probability that both are blue?

- A) $1/15$
- B) $1/9$
- C) $2/15$
- D) $1/3$

134. A right triangle has legs of length 6 units and 8 units. What is its area?

- A) 14 square units
- B) 24 square units
- C) 20 square units
- D) 48 square units

135. Each exterior angle of a regular polygon measures 45° . How many sides does the polygon have?

- A) 6
- B) 8
- C) 7
- D) 9

136. A container holds 2.5 liters. If 1 liter = 1,000 milliliters, how many 250-milliliter cups can be filled completely?

- A) 8
- B) 9
- C) 10
- D) 12

137. A map uses a scale of 1 inch = 18 miles. Two towns are $3\frac{1}{2}$ inches apart on the map. What is the actual distance?

- A) 54 miles
- B) 60 miles
- C) 63 miles
- D) 72 miles

138. A tank is $\frac{3}{5}$ full. After 24 liters are added, it is $\frac{9}{10}$ full. What is the tank's total capacity?

- A) 60 liters
- B) 72 liters
- C) 80 liters
- D) 96 liters

139. What is the value of $2.4 \div 0.06$?

- A) 4
- B) 40
- C) 400
- D) 0.4

140. A store reduces a \$75 jacket by 20%, then applies a \$6 coupon to the reduced price. What is the final price before tax?

- A) \$48
- B) \$51
- C) \$57
- D) \$54

141. For the function $y = 3x - 4$, which value of x produces $y = 17$?

- A) 7
- B) 6
- C) 5
- D) 8

142. The first four terms of a sequence are 5, 8, 14, 23. The amount added increases by 3 each time. What is the next term?

- A) 29
- B) 32
- C) 38
- D) 35

143. The mean of five quiz scores is 84. Four of the scores are 78, 81, 86, and 90. What is the fifth score?

- A) 82
- B) 83
- C) 85
- D) 87

144. A bag contains 4 red, 3 blue, and 5 yellow tiles. One tile is selected, not replaced, and then a second tile is selected. What is the probability that both tiles are blue?

- A) $\frac{1}{16}$
- B) $\frac{1}{12}$
- C) $\frac{1}{22}$
- D) $\frac{1}{6}$

145. A bus recorded these passenger counts on five trips: 18, 24, 21, 27, and 20. A sixth trip raised the mean for all six trips to 23. How many passengers were on the sixth trip?

- A) 28
- B) 30
- C) 26
- D) 32

146. Two interior angles of a triangle measure 48° and 67° . What is the measure of the exterior angle at the third vertex?

- A) 65°
- B) 115°
- C) 125°
- D) 133°

147. A circle has a diameter of 14 centimeters. Using $\pi = \frac{22}{7}$, what is its circumference?

- A) 22 centimeters
- B) 28 centimeters
- C) 44 centimeters
- D) 154 centimeters

148. A right triangle has legs of 9 units and 12 units. What is the length of its hypotenuse?

- A) 15 units
- B) 18 units
- C) 21 units
- D) 25 units

149. A recipe uses 350 milliliters of broth for one batch. How many liters of broth are needed for 6 batches? Use 1,000 milliliters = 1 liter.

- A) 1.75 liters
- B) 2.1 liters
- C) 2.6 liters
- D) 3.5 liters

150. A movie begins at 6:47 p.m. and lasts 1 hour 56 minutes. At what time does it end?

- A) 8:33 p.m.
- B) 8:37 p.m.
- C) 8:53 p.m.
- D) 8:43 p.m.

Section 4: Ability

50 questions | Recommended time: 35 minutes

151. Open shapes: ○ □ △ ◇ ☆

Filled shapes: ● ■ ▲ ◆ ★

Complete the 2 × 2 matrix.

Top-left: ○

Top-right: ●●

Bottom-left: □□

Bottom-right: ?

In each row, the shape changes from open to filled and one shape is added. In each column, circles change to squares without changing the count.

A) ■■■

B) □□□

C) ●●

D) ■■

152. Complete the 2 × 2 matrix.

Top-left: ↑ ○

Top-right: → ○

Bottom-left: ↑ □

Bottom-right: ?

Across each row, the arrow rotates 90° clockwise. Down each column, the circle changes to a square.

A) ← □

B) → □

C) → ○

D) ↓ □

153. Complete the 3 × 3 matrix.

Row 1: ○ | □ | ○□

Row 2: △ | ◇ | △◇

Row 3: ☆ | ★ | ?

In each row, the third cell combines the figures from the first two cells in the same order.

A) ★☆

B) ☆

C) ☆★

D) ★

154. Complete the 2×2 matrix.

Top-left:

Top: ●

Bottom: ○

Top-right:

Top: ○

Bottom: ●

Bottom-left:

Top: ■

Bottom: □

Bottom-right: ?

Across each row, the top and bottom positions exchange. Down each column, circles change to squares while fill remains the same.

A) Top: ■; Bottom: □

B) Top: ●; Bottom: ○

C) Top: ○; Bottom: ●

D) Top: □; Bottom: ■

155. Complete the 3×3 matrix.

Row 1: ○ | ○○ | ○○○

Row 2: □□ | □□□ | □□□□

Row 3: △△△ | △△△△ | ?

Across each row, one shape is added from left to right. Down each column, the starting count increases by one.

A) △△△△△

B) △△△

C) ▲▲▲▲▲

D) △△△△△△

156. A square sheet is folded in half from left to right. One circular hole is punched near the upper-left corner of the folded rectangle, away from the fold. When the sheet is completely unfolded, which pattern appears?

A) Two holes near the upper-left corner

B) Two holes near the upper-left and upper-right corners

C) Two holes near the upper-right and lower-right corners

D) Four holes, one near each corner

157. A square sheet is folded in half from top to bottom, then in half from right to left. One hole is punched near the lower-right corner of the final folded square, away from both folds. What appears when the sheet is completely unfolded?

A) Two holes aligned horizontally

B) Two holes aligned vertically

C) Four holes placed symmetrically across both centerlines

D) Eight holes placed around the edges

158. A rectangular sheet is folded in half along its vertical centerline. A small triangular cut is made on the folded edge at the exact vertical center of the folded rectangle. What appears when the sheet is unfolded?

A) Two separate triangular cuts on the outer edges

B) One triangular cut on the left edge

C) Four triangular cuts, one near each corner

D) One diamond-shaped opening centered on the vertical centerline

159. A square sheet is folded along the diagonal from upper-left to lower-right. One hole is punched near the upper-left corner of the folded triangle, but not on the fold. Which statement describes the unfolded result?

- A) Two holes mirrored across the upper-left-to-lower-right diagonal
- B) Two holes mirrored across the other diagonal
- C) Four holes mirrored across both diagonals
- D) One hole remains because the punch was near a corner

160. The three groups below share the same rule.

Group 1: ○ ● ○

Group 2: □ ■ □

Group 3: △ ▲ △

Which choice follows the same rule?

- A) ○ ● ●
- B) ◇ ◆ ◇
- C) ☆ ☆ ★
- D) ■ □ ■

161. The three arrangements below share the same rule.

Arrangement 1: ↑ → ↓

Arrangement 2: → ↓ ←

Arrangement 3: ↓ ← ↑

Which choice follows the same rule?

- A) ↑ ↓ →
- B) ← ↓ →
- C) ← ↑ →
- D) → ↑ ←

162. The three figures below share the same rule.

Figure 1: ○ inside □

Figure 2: △ inside ◇

Figure 3: ☆ inside ○

Which choice follows the same rule?

- A) ● inside □
- B) ■ inside ◆
- C) ▲ inside △
- D) ◇ inside ☆

163. Complete the 2 × 2 matrix.

Top-left: ○ above -

Top-right: - with ○ below

Bottom-left: □ on the left of |

Bottom-right: ?

Across each row, the shape moves to the opposite side of the line. Down each column, ○ changes to □ and - rotates to |.

- A) | with □ on the left
- B) | with □ on the right
- C) - with □ below
- D) | with ○ on the right

164. Complete the 3×3 matrix.

Row 1: ○ | □ | △

Row 2: □ | △ | ○

Row 3: △ | ○ | ?

Each row and column contains one circle, one square, and one triangle.

- A) ○
- B) □
- C) ◇
- D) △

165. Open shapes: ○ □ △

Filled shapes: ● ■ ▲

Complete the 2×2 matrix.

Top-left: ○□

Top-right: ●□

Bottom-left: ○△

Bottom-right: ?

Across each row, only the first shape changes from open to filled. Down each column, only the second shape changes from square to triangle.

- A) ●△
- B) ▲○
- C) ○▲
- D) ●□

166. Complete the 3×3 matrix.

Row 1: ↑ | → | ↓

Row 2: → | ↓ | ←

Row 3: ↓ | ← | ?

Across each row and down each column, the arrow rotates 90° clockwise from one cell to the next.

- A) ←
- B) ↓
- C) ↑
- D) →

167. A square sheet is folded in half from bottom to top. One hole is punched near the lower-left corner of the folded rectangle, away from the fold. When the sheet is unfolded, which pattern appears?

- A) Two holes near the upper-left and lower-left corners
- B) Two holes near the upper-left and upper-right corners
- C) Four holes, one near each corner
- D) Two holes near the lower-left and lower-right corners

168. A square sheet is folded along the diagonal from upper-right to lower-left. A hole is punched near the lower-left corner of the folded triangle, away from the fold. What appears when the sheet is unfolded?

- A) Four holes symmetric across both diagonals
- B) One hole on the diagonal
- C) Two holes mirrored across the upper-right-to-lower-left diagonal
- D) Two holes mirrored across the horizontal centerline

169. A square sheet is folded from left to right and then from bottom to top. A hole is punched at the exact center of the final folded square, away from both folded edges. What pattern appears after complete unfolding?

- A) Two holes on the vertical centerline
- B) Eight holes around the border
- C) One hole at the center of the original sheet
- D) Four holes, one in each quadrant, symmetric across both centerlines

170. A rectangular sheet is folded in half from right to left. A small semicircular cut is made at the midpoint of the folded rectangle's top outer edge, away from the fold. What appears along the top edge when the sheet is unfolded?

- A) One circular opening at the center
- B) Two matching semicircular notches, one in each half of the top edge
- C) Four semicircular notches along the top edge
- D) One semicircular notch at the left corner

171. The three arrangements below share the same rule.

Arrangement 1: ○ □ ○ □

Arrangement 2: △ ◇ △ ◇

Arrangement 3: ☆ + ☆ +

Which choice follows the same rule?

- A) ○ ○ □ □
- B) ▲ ▲ ▲ ▲
- C) ◆ □ ◆ □
- D) + × + ×

172. The three figures below share the same rule.

Figure 1: Top: ○; Center: □; Bottom: ○

Figure 2: Top: △; Center: ◇; Bottom: △

Figure 3: Top: ☆; Center: +; Bottom: ☆

Which choice follows the same rule?

- A) Top: □; Center: ○; Bottom: □
- B) Top: ○; Center: □; Bottom: △
- C) Top: ●; Center: □; Bottom: ○
- D) Top: +; Center: +; Bottom: ×

173. The three arrangements below share the same rule.

Arrangement 1: ○ inside □; ● outside □

Arrangement 2: △ inside ◇; ▲ outside ◇

Arrangement 3: ☆ inside ○; ★ outside ○

Which choice follows the same rule?

- A) ■ inside □; □ outside □
- B) ◆ inside △; ◇ outside △
- C) ◇ inside ☆; ◆ outside ☆
- D) ▲ inside ○; ▲ outside ○

174. The three sequences below share the same rule.

Sequence 1: $\uparrow \uparrow \rightarrow$

Sequence 2: $\rightarrow \rightarrow \downarrow$

Sequence 3: $\downarrow \downarrow \leftarrow$

Which choice follows the same rule?

- A) $\leftarrow \uparrow \uparrow$
- B) $\uparrow \rightarrow \rightarrow$
- C) $\leftarrow \downarrow \leftarrow$
- D) $\leftarrow \leftarrow \uparrow$

175. Complete the matrix. In each row, the third cell contains the shapes found in both of the first two cells, while shapes appearing in only one cell are removed.

Row 1: $\circ + \mid \circ \times \mid \circ$

Row 2: $\square \triangle \mid \triangle \diamond \mid \triangle$

Row 3: $\star \blacklozenge \mid \blacklozenge + \mid ?$

- A) $\star$
- B) $+$
- C) $\blacklozenge$
- D) $\star \blacklozenge +$

176. Complete the matrix. Moving from left to right, the arrow rotates 90° clockwise and the shape changes from open to filled, then back to open.

Row 1: $\circ \uparrow \mid \bullet \rightarrow \mid \circ \downarrow$

Row 2: $\square \leftarrow \mid \blacksquare \uparrow \mid \square \rightarrow$

Row 3: $\triangle \downarrow \mid \blacktriangle \leftarrow \mid ?$

- A) $\triangle \uparrow$
- B) $\blacktriangle \uparrow$
- C) $\triangle \rightarrow$
- D) $\blacktriangle \rightarrow$

177. Complete the matrix. In each row, the number of shapes in the third cell equals the total in the first two cells; the shape type alternates by row.

Row 1: $\circ \mid \circ \circ \mid \circ \circ \circ$

Row 2: $\square \square \mid \square \mid \square \square \square$

Row 3: $\triangle \triangle \mid \triangle \triangle \triangle \mid ?$

- A) $\triangle \triangle \triangle \triangle$
- B) $\triangle \triangle \triangle \triangle \triangle$
- C) $\blacktriangle \blacktriangle \blacktriangle \blacktriangle \blacktriangle$
- D) $\triangle \triangle \triangle \triangle \triangle \triangle$

178. Complete the matrix. From the first cell to the second, the two symbols exchange positions. From the second to the third, both symbols change from open to filled.

Row 1: $\circ \square \mid \square \circ \mid \blacksquare \bullet$

Row 2: $\triangle \diamond \mid \diamond \triangle \mid \blacklozenge \blacktriangle$

Row 3: $\star \circ \mid \circ \star \mid ?$

- A) $\star \bullet$
- B) $\circ \star$
- C) $\bullet \star$
- D) $\bullet \blackstar$

179. Complete the matrix. Down each column, the arrow in the middle cell points opposite the arrow in the top cell, and the bottom arrow points 90° clockwise from the middle arrow.

Column 1: ↑, ↓, ←

Column 2: →, ←, ↑

Column 3: ↓, ↑, ?

- A) ←
- B) ↑
- C) →
- D) ↓

180. A square sheet is folded left to right along the vertical centerline. It is then folded bottom to top along the horizontal centerline. One hole is punched in the upper-right corner of the folded square. After the sheet is completely unfolded, where are the holes?

- A) At the centers of the top and bottom edges
- B) At all four corners
- C) At the midpoints of all four edges
- D) In four positions surrounding the center

181. A rectangular sheet is folded top to bottom along its horizontal centerline. A hole is punched on the folded rectangle's vertical centerline, halfway between its top and bottom edges. After unfolding, what pattern appears?

- A) Two holes on the vertical centerline, equally spaced above and below the horizontal centerline
- B) Two holes on the horizontal centerline, one left and one right
- C) Four holes, one in each quadrant
- D) One hole at the center

182. A square sheet is folded along the diagonal from upper-left to lower-right. One hole is punched in the folded triangle near the midpoint of the left edge, not on the fold. After unfolding, what must be true?

- A) There will be one hole on the diagonal.
- B) There will be two holes reflected across the diagonal.
- C) There will be four holes around the center.
- D) There will be two holes reflected across the vertical centerline.

183. A square sheet is folded right to left along the vertical centerline. The folded rectangle is then folded top to bottom. A triangular cut is made in the corner where the two folded edges meet. After complete unfolding, how many matching triangular cuts appear around the sheet's center?

- A) 2
- B) 3
- C) 4
- D) 8

184. The three figures ○●○, ◻■◻, and △▲△ share a rule. Which choice follows the same rule?

- A) ◇◆◇
- B) ◆◇◆
- C) ☆☆☆
- D) ★☆☆

185. The three arrangements below share a rule:

1. Top: ○; Bottom-left: □; Bottom-right: □
 2. Top: △; Bottom-left: ◇; Bottom-right: ◇
 3. Top: ☆; Bottom-left: ○; Bottom-right: ○
- Which choice follows the same rule?**

- A) Top: ■; Bottom-left: □; Bottom-right: ■
- B) Top: ◆; Bottom-left: △; Bottom-right: △
- C) Top: ○; Bottom-left: □; Bottom-right: △
- D) Top: ▲; Bottom-left: ▲; Bottom-right: ▲

186. The groups $\uparrow \rightarrow \downarrow$, $\rightarrow \downarrow \leftarrow$, and $\downarrow \leftarrow \uparrow$ share a rule. Which choice follows the same rule?

- A) $\uparrow \downarrow \rightarrow$
- B) $\leftarrow \downarrow \rightarrow$
- C) $\rightarrow \leftarrow \downarrow$
- D) $\leftarrow \uparrow \rightarrow$

187. The three figures ○|○, □-□, and △/△ share a rule. Which choice follows the same rule?

- A) ◇\◆
- B) ●|○
- C) ◆-◇
- D) ☆+☆

188. Complete the matrix. In each row, the shape moves one position clockwise around the four labeled positions, while its open or filled form stays the same.

Row 1: Top-left: ○ | Top-right: ○ | Bottom-right: ○

Row 2: Bottom-left: ■ | Top-left: ■ | Top-right: ■

Row 3: Bottom-right: △ | Bottom-left: △ | ?

- A) Top-left: △
- B) Top-right: △
- C) Bottom-left: △
- D) Bottom-right: △

189. Complete the matrix. In each row, the third cell places the first cell's shape inside the second cell's shape.

Row 1: ○ | □ | ○ inside □

Row 2: △ | ◇ | △ inside ◇

Row 3: ☆ | ○ | ?

- A) ○ inside ☆
- B) ☆ beside ○
- C) ☆ inside ○
- D) ★ inside ●

190. Complete the matrix. Across each row, the outer shapes stay in place, and the center alternates open, filled, open.

Row 1: ○ □ ○ | ○ ■ ○ | ○ □ ○

Row 2: △ ◇ △ | △ ◆ △ | △ ◇ △

Row 3: ☆ ○ ☆ | ☆ ● ☆ | ?

- A) ★ ○ ★
- B) ☆ ● ☆
- C) ★ ● ★
- D) ☆ ○ ☆

191. Complete the matrix. From top to bottom in each column, one new line is added to the right of the shape.

Column 1: ○, ○|, ○||

Column 2: □|, □||, □|||

Column 3: △||, △|||, ?

- A) △||||
- B) △|||
- C) △||
- D) △|

192. A square sheet is folded from left to right along the vertical centerline. One hole is punched in the folded rectangle near its upper-left corner, not on an edge or fold. After unfolding, where are the holes?

- A) Two holes in the upper half, reflected across the vertical centerline
- B) Two holes in the left half, reflected across the horizontal centerline
- C) Four holes, one in each quadrant
- D) One hole on the vertical centerline

193. A rectangular sheet is folded bottom to top along the horizontal centerline. It is then folded from left to right along the vertical centerline. A hole is punched exactly on the vertical folded edge but away from the horizontal folded edge. After complete unfolding, how many holes appear?

- A) 1
- B) 4
- C) 8
- D) 2

194. A square sheet is folded along the diagonal from upper-right to lower-left. A small semicircular cut is made on the diagonal folded edge, away from both corners. What appears when the sheet is unfolded?

- A) Two separate semicircular cuts reflected across the diagonal
- B) One complete circular hole centered on the diagonal
- C) Four semicircular cuts around the center
- D) One semicircular cut on the outer edge

195. A square sheet is folded top to bottom along the horizontal centerline. The folded rectangle is then folded right to left along the vertical centerline. One hole is punched in the lower-left corner of the final folded square, which is an original corner of the sheet and lies on neither fold. After complete unfolding, where are the holes?

- A) At two original corners
- B) In four positions surrounding the center
- C) At the midpoint of each outer edge
- D) At all four original corners

196. The groups ○|●, □|■, and △|▲ share a rule. Which choice follows the same rule?

- A) ◆|◇
- B) ☆|☆
- C) ◇|◆
- D) ★|☆

197. The arrangements below share a rule:

1. Outer: □; **Inner:** ○; **symbol below:** ○

2. Outer: ◇; **Inner:** △; **symbol below:** △

3. Outer: ○; **Inner:** ☆; **symbol below:** ☆

Which choice follows the same rule?

A) Outer: △; Inner: □; symbol below: ○

B) Outer: ☆; Inner: ◇; symbol below: ◇

C) Outer: □; Inner: ●; symbol below: ○

D) Outer: ◇; Inner: △; symbol below: ◇

198. The groups $\uparrow \rightarrow \downarrow \leftarrow$, $\rightarrow \downarrow \leftarrow \uparrow$, and $\downarrow \leftarrow \uparrow \rightarrow$ share a rule. Which choice follows the same rule?

A) $\leftarrow \uparrow \rightarrow \downarrow$

B) $\uparrow \downarrow \rightarrow \leftarrow$

C) $\rightarrow \uparrow \leftarrow \downarrow$

D) $\leftarrow \downarrow \rightarrow \uparrow$

199. The figures ○+○, □×□, and △|△ share a rule. Which choice follows the same rule?

A) ◇◇+

B) +☆☆

C) ◆-◆

D) ☆/☆

200. The arrangements below share a rule:

1. Top: ○; **Center:** □; **Bottom:** ○

2. Top: △; **Center:** ◇; **Bottom:** △

3. Top: ☆; **Center:** ○; **Bottom:** ☆

Which choice follows the same rule?

A) Top: □; Center: □; Bottom: ○

B) Top: ◆; Center: △; Bottom: ◇

C) Top: ○; Center: ☆; Bottom: ○

D) Top: ▲; Center: ▲; Bottom: △

Answer Key

Question numbers continue across all five sections.

Reading

1. A	2. B	3. C	4. D	5. A	6. B	7. C	8. D
9. A	10. B	11. C	12. D	13. A	14. A	15. B	16. C
17. D	18. B	19. D	20. A	21. C	22. D	23. B	24. A
25. C	26. A	27. B	28. C	29. D	30. B	31. A	32. C
33. D	34. A	35. B	36. C	37. A	38. D	39. C	40. A
41. D	42. B	43. A	44. C	45. B	46. D	47. C	48. A
49. B	50. D						

Written Expression

51. B	52. A	53. C	54. D	55. B	56. C	57. A	58. D
59. B	60. C	61. A	62. D	63. B	64. A	65. B	66. C
67. D	68. C	69. A	70. B	71. A	72. C	73. D	74. C
75. B	76. D	77. C	78. A	79. B	80. D	81. A	82. B
83. C	84. D	85. B	86. A	87. D	88. C	89. B	90. A
91. A	92. D	93. B	94. C	95. A	96. D	97. B	98. C
99. C	100. D						

Mathematics

101. A	102. B	103. C	104. D	105. A	106. B	107. C	108. D
109. A	110. B	111. C	112. D	113. D	114. A	115. B	116. C
117. D	118. B	119. A	120. D	121. C	122. A	123. B	124. C
125. D	126. A	127. C	128. B	129. D	130. A	131. D	132. D
133. A	134. B	135. B	136. C	137. C	138. C	139. B	140. D
141. A	142. D	143. C	144. C	145. A	146. B	147. C	148. A
149. B	150. D						

Ability

151. A	152. B	153. C	154. D	155. A	156. B	157. C	158. D
159. A	160. B	161. C	162. D	163. B	164. B	165. A	166. C
167. A	168. C	169. D	170. B	171. D	172. A	173. C	174. D
175. C	176. A	177. B	178. D	179. C	180. B	181. A	182. B
183. C	184. A	185. B	186. D	187. D	188. A	189. C	190. D
191. B	192. A	193. D	194. B	195. D	196. C	197. B	198. A

199. D	200. C						
--------	--------	--	--	--	--	--	--

Detailed Explanations

Reading

1. TACHS-R1-001 - Answer A: To observe details that could improve the sign
Mara studies the garden's vines, leaves, and angles before revising the sign's design.
2. TACHS-R1-002 - Answer B: She cares about how her work makes people feel.
Mara wants the sign to feel welcoming, not merely readable.
3. TACHS-R1-003 - Answer C: Observation and flexibility can turn an imperfection into a strength.
Mara draws inspiration from the garden and transforms an accidental paint drop into an appealing bee.
4. TACHS-R1-004 - Answer D: formal and lacking liveliness.
The office-hallway comparison shows that the letters appear overly formal rather than warm and lively.
5. TACHS-R1-005 - Answer A: To learn how to use specialized equipment safely
The passage states that some borrowers must complete a safety lesson before using specialized equipment.
6. TACHS-R1-006 - Answer B: encourage people to share skills as well as objects.
The sewing-machine example connects borrowing an object with learning repair skills from a volunteer.
7. TACHS-R1-007 - Answer C: It explains a concept, describes its benefits, and acknowledges its challenges.
The passage defines the program, presents economic and environmental benefits, and then discusses operational demands.
8. TACHS-R1-008 - Answer D: possible to handle successfully.
Supporters accept that planning challenges exist but believe they can be dealt with effectively.
9. TACHS-R1-009 - Answer A: Sustained observation can deepen understanding in museums and everyday life.
The passage argues that slow, careful attention reveals details and patterns across many settings.
10. TACHS-R1-010 - Answer B: provide context after viewers have first made their own observations.
The author says labels can add context but should not replace a viewer's initial observation.
11. TACHS-R1-011 - Answer C: To encourage readers to practice deliberate, evidence-based attention
The author offers questions and examples designed to promote slower, more thoughtful observation.
12. TACHS-R1-012 - Answer D: Where did my eyes land first?
This question appears directly among the simple prompts recommended for slow looking.
13. TACHS-R1-013 - Answer A: To show that careful attention is useful in settings beyond art
The examples extend the value of close observation from museums to everyday and academic contexts.
14. TACHS-R1-014 - Answer A: A pad beneath one of Jonah's clarinet keys is torn.
Ms. Alvarez examines the clarinet and finds a torn pad beneath one key.
15. TACHS-R1-015 - Answer B: He is encouraging Priya to focus and continue.
Jonah's nod directs the anxious violinist back toward the conductor and helps her resume.
16. TACHS-R1-016 - Answer C: Responding calmly to an unexpected problem can strengthen confidence.
Jonah adapts to an unfamiliar instrument and learns that preparation includes handling change.
17. TACHS-R1-017 - Answer D: required extra effort to press.
The unfamiliar spare clarinet's keys feel harder for Jonah's fingers to operate.
18. TACHS-R1-018 - Answer B: Gravity
The passage states that gravity pulls drops heavy enough to move into the channel.
19. TACHS-R1-019 - Answer D: To clarify a limitation and prevent an exaggerated impression of the technology
The statement emphasizes that collectors capture existing moisture and cannot provide unlimited water.
20. TACHS-R1-020 - Answer A: It explains how a device works, identifies factors affecting it, and discusses its practical limits.
The paragraphs move from operation to design conditions and then to uses and limitations.
21. TACHS-R1-021 - Answer C: appropriate for the purpose.
Suitable conditions are the fog and wind conditions appropriate for effective collection.
22. TACHS-R1-022 - Answer D: Recorded the signal and repeated it for the next station
Operators wrote down incoming symbols before relaying them to the following station.
23. TACHS-R1-023 - Answer B: appreciative of its ingenuity while aware of its weaknesses.
The author highlights the network's speed and principle while acknowledging weather, error, and expense.

24. TACHS-R1-024 - Answer A: A chain of visual stations sped communication by relaying coded signals, although weather and human error limited the system.
This choice includes both the system's central method and its principal limitations.
25. TACHS-R1-025 - Answer C: It helped detect an incorrectly received group of symbols before the message continued.
Waiting for confirmation allowed an operator to verify each symbol group before forwarding more.
26. TACHS-R1-026 - Answer A: The weight beneath the hull had shifted.
Mara finds the off-center metal weight after the boat circles.
27. TACHS-R1-027 - Answer B: She is willing to investigate her own mistake.
Mara resists an easy excuse and checks her rushed work.
28. TACHS-R1-028 - Answer C: To emphasize the practical lessons Mara gained
The entries record specific improvements learned through testing.
29. TACHS-R1-029 - Answer D: would expose weaknesses more readily
Unlike the bathtub, the windy pond reveals flaws in the boat.
30. TACHS-R1-030 - Answer B: A note from the eggbeater's former owner
The storage box contains the donor's memory of Sunday baking.
31. TACHS-R1-031 - Answer A: He believes the description lacks a human connection.
The mechanical explanation does not show why the object mattered to people.
32. TACHS-R1-032 - Answer C: Brief personal details can make historical objects meaningful.
The added memory prompts visitors to connect the object with their experiences.
33. TACHS-R1-033 - Answer D: It demonstrates the effect of the sentence Luis added.
The label inspires a grandfather to share his own memory.
34. TACHS-R1-034 - Answer A: Competition for light encourages upward growth.
The first paragraph directly connects close planting with rapid upward growth.
35. TACHS-R1-035 - Answer B: To separate the forest's effects from other local conditions
Similar unplanted areas provide a comparison for judging the planting's actual effects.
36. TACHS-R1-036 - Answer C: Favorable but carefully qualified
The author presents benefits while acknowledging limits, maintenance, and the need for research.
37. TACHS-R1-037 - Answer A: Tiny forests are dense urban plantings with potential benefits that require careful planning and long-term study.
This choice includes the method, possible value, and need for evaluation.
38. TACHS-R1-038 - Answer D: pressed tightly together
Loosening compacted soil allows roots to spread through soil that had been compressed.
39. TACHS-R1-039 - Answer C: He was avoiding a loose strip of tape.
The passage directly states that Mateo paused to avoid loose tape on the floor.
40. TACHS-R1-040 - Answer A: She should observe the entire situation before deciding what caused a problem.
Nia solves the mismatch only after watching the dancers instead of assuming her drumming caused it.
41. TACHS-R1-041 - Answer D: Nia discovers that an apparent timing error is caused by an obstacle affecting the dancers.
The summary includes the timing problem, its actual cause, and Nia's discovery.
42. TACHS-R1-042 - Answer B: directing
The rhythm directs or leads the dancers' movement through the routine.
43. TACHS-R1-043 - Answer A: Its particles contain large spaces between them.
The passage says water passes quickly through the relatively large spaces in sandy soil.
44. TACHS-R1-044 - Answer C: To explain how soil structure affects water movement and how that structure can be improved
The passage explains soil pathways and concludes with practical ways to protect or improve them.
45. TACHS-R1-045 - Answer B: It describes factors affecting water movement, then gives ways to protect or improve soil structure.
The first paragraphs explain soil, organic matter, and worms before the final paragraph offers gardening practices.
46. TACHS-R1-046 - Answer D: tightly packed
In context, dense clay has small spaces and is therefore tightly packed.
47. TACHS-R1-047 - Answer C: To check unusual results that images or software may produce
The passage states that plants, shadows, and waves can affect aerial results, so Ortiz checks them in person.

48. TACHS-R1-048 - Answer A: It is most useful when combined with observation and human judgment.
Ortiz uses aerial tools but continues field checks and values community observations.

49. TACHS-R1-049 - Answer B: respectful and thoughtful

The author presents Ortiz's careful methods and broader community work with clear approval.

50. TACHS-R1-050 - Answer D: Ortiz combines evolving tools, field checks, and community participation to study coastal change.

This choice captures all three major parts of Ortiz's approach described throughout the passage.

Written Expression

51. TACHS-WE1-001 - Answer B: Sentence 5

The drama-club performance is unrelated to the proposed trees and their schoolyard benefits.

52. TACHS-WE1-002 - Answer A: Our environmental club proposes adding three shade trees along the paved area's western edge.

Replacing "its" with "the paved area's" removes the unclear pronoun reference directly and concisely.

53. TACHS-WE1-003 - Answer C: In addition,

Sentence 4 adds a second benefit of planting trees, so an additive transition is appropriate.

54. TACHS-WE1-004 - Answer D: Because native species suit local conditions, club members can help water them, while a grounds worker supervises long-term care.

This choice joins the related ideas with correct subordination and coordination.

55. TACHS-WE1-005 - Answer B: adjusts the seasonings according to the flavor.

"Adjusts" and "according to the flavor" state the action and its basis precisely.

56. TACHS-WE1-006 - Answer C: After sentence 6

"These visual descriptions" clearly refers to the sauce's appearance described in sentence 6.

57. TACHS-WE1-007 - Answer A: Recording the recipes required patience, but it preserved knowledge that might otherwise have been lost.

The coordinating conjunction "but" correctly joins two independent clauses and clarifies their contrast.

58. TACHS-WE1-008 - Answer D: By translating experience into clear directions, I created a guide that future cooks can actually use.

The sentence reinforces both the recording process and its practical value, effectively concluding the passage.

59. TACHS-WE1-009 - Answer B: Neither the coaches nor the captain was available for the interview.

With "neither...nor," the verb agrees with the nearer singular subject "captain."

60. TACHS-WE1-010 - Answer C: The workshop teaches students to plan a budget, compare prices, and save money.

All three items use parallel base-form verbs: "plan," "compare," and "save."

61. TACHS-WE1-011 - Answer A: The science teacher asked Maya and me to organize the samples.

"Maya and me" correctly functions as the object of the verb "asked."

62. TACHS-WE1-012 - Answer D: Line 4

The singular subject "group" requires "has reached," not "have reached."

63. TACHS-WE1-013 - Answer B: Line 2

The object pronoun "us," not the subject pronoun "we," must follow "showed."

64. TACHS-WE1-014 - Answer A: Sentence 6

The council members' reading preferences do not help explain the study or its recommendations.

65. TACHS-WE1-015 - Answer B: First, volunteers used sound meters to record noise in six areas on three separate days.

This revision precisely identifies the tool, measurement, locations, and schedule.

66. TACHS-WE1-016 - Answer C: After reviewing the results, the council recommended adding two small tables and posting quiet-zone signs in one section of the courtyard.

The past-tense verb and parallel gerunds correctly describe the council's completed recommendation.

67. TACHS-WE1-017 - Answer D: Although the library corner was consistently quieter, the covered courtyard offered thirty seats instead of only eight.

The subordinating conjunction clearly expresses contrast without creating a fragment or run-on.

68. TACHS-WE1-018 - Answer C: One rainy Saturday, our class learned how planting and later monitoring can protect an eroded stream bank.

The sentence introduces the setting and both major stages developed in the passage.

69. TACHS-WE1-019 - Answer A: Although the work was muddy, everyone stayed focused until each plant was secure.

"Although" correctly shows contrast and joins the ideas in one complete sentence.

70. TACHS-WE1-020 - Answer B: Next spring, the group will return to measure the bank's remaining plant cover.

This version preserves the meaning while eliminating wordiness and repetition.

71. TACHS-WE1-021 - Answer A: The selected grasses also tolerate the stream bank's usual moisture and sunlight.

This detail directly supports why those native grasses were appropriate for the site.

72. TACHS-WE1-022 - Answer C: Neither the captain nor the players were prepared for the schedule change.

With "neither...nor," the verb agrees with the nearer plural subject "players."

73. TACHS-WE1-023 - Answer D: The science teacher asked Maya and me to organize the samples.
"Me" is the correct objective pronoun after the verb "asked."
74. TACHS-WE1-024 - Answer C: Line 3
The past-tense context requires "gave," not the present-tense form "give."
75. TACHS-WE1-025 - Answer B: Line 2
A comparison between two plans requires the comparative "more practical," not the superlative "most practical."
76. TACHS-WE1-026 - Answer D: Line 4
The adverb "slowly," rather than the adjective "slow," must modify "poured."
77. TACHS-WE1-027 - Answer C: Sentence 5
The cafeteria's soup does not support the proposal for a repair station.
78. TACHS-WE1-028 - Answer A: Students could borrow simple tools for small repairs, with adult supervision and instruction cards explaining safe use.
This revision combines both ideas clearly and grammatically.
79. TACHS-WE1-029 - Answer B: a brief fix could extend their useful life
The revision is specific, concise, and appropriate for a proposal.
80. TACHS-WE1-030 - Answer D: A monthly record could show how many items were repaired instead of discarded.
Repair data would provide relevant evidence for the station's waste-reduction goal.
81. TACHS-WE1-031 - Answer A: location
"Location" precisely identifies where each observation occurred.
82. TACHS-WE1-032 - Answer B: Although the information does not answer every question about frog populations, it helps researchers notice changes that may deserve closer study.
The subordinating conjunction clearly expresses the contrast without creating an error.
83. TACHS-WE1-033 - Answer C: Repeated checks of the same ponds allow scientists to compare observations across years.
This revision preserves the meaning while removing unnecessary wording.
84. TACHS-WE1-034 - Answer D: Neither the coaches nor the captain was prepared to cancel practice.
With "neither...nor," the verb agrees with the nearer singular subject "captain."
85. TACHS-WE1-035 - Answer B: me
The pronoun is the object of "gave," so the objective form "me" is required.
86. TACHS-WE1-036 - Answer A: The workshop taught us to measure carefully, to cut safely, and to assemble patiently.
All three items use matching infinitive phrases.
87. TACHS-WE1-037 - Answer D: Line 4
The singular subject "neither hiker" requires "has reached," not "have reached."
88. TACHS-WE1-038 - Answer C: Line 3
After "watched an employee," the verb should be "stack" or "stacking," not "stacked."
89. TACHS-WE1-039 - Answer B: Sentence 4
The basketball uniform is not developed or connected to the passage's explanation of designing the exhibit.
90. TACHS-WE1-040 - Answer A: Because the hallway is narrow, we tested the display before attaching anything to the walls, revealing where people could stop without blocking the doors.
This revision clearly joins the cause, action, and result in one complete sentence.
91. TACHS-WE1-041 - Answer A: Each section presented photographs, a brief caption, and one recorded memory.
The revision removes unnecessary words while preserving every important detail.
92. TACHS-WE1-042 - Answer D: A graduate recalled attending public lectures there before the school added a separate community room.
This detail directly supports the auditorium's former use as a neighborhood meeting place.
93. TACHS-WE1-043 - Answer B: This warning should be given soon enough to prevent the walker from being startled.
The sentence uses clear, precise language and an appropriately formal tone.
94. TACHS-WE1-044 - Answer C: These simple habits make a busy trail more predictable and comfortable for everyone.
This option combines both ideas in a concise, grammatically complete sentence.
95. TACHS-WE1-045 - Answer A: Dog owners should also move aside when stopping, especially near curves or narrow bridges.
Replacing "They" with "Dog owners" removes uncertainty about the pronoun's antecedent.

96. TACHS-WE1-046 - Answer D: Of the two routes, the forest path is shorter.
The comparative form "shorter" correctly compares two routes without a redundant modifier.
97. TACHS-WE1-047 - Answer B: The science teacher gave Elena and me permission to use the lab.
"Elena and me" correctly functions as the compound object of "gave."
98. TACHS-WE1-048 - Answer C: Line 3
The singular subject "group" requires the singular verb "is preparing."
99. TACHS-WE1-049 - Answer C: Line 3
The present-tense verb "discovers" improperly shifts from the passage's established past tense.
100. TACHS-WE1-050 - Answer D: Line 4
"Returned back" is redundant because "returned" already means went back.

Mathematics

101. TACHS-M1-001 - Answer A: 60 liters

The added 18 liters equal $9/10 - 3/5 = 3/10$ of capacity, so $18 \div 3/10 = 60$.

102. TACHS-M1-002 - Answer B: 15

Evaluate parentheses first: $5 + 1 = 6$; then $3 \times 6 \div 2 = 9$, and $24 - 9 = 15$.

103. TACHS-M1-003 - Answer C: 72%

After both discounts, the price is $0.80 \times 0.90 = 0.72$, or 72% of the original.

104. TACHS-M1-004 - Answer D: 19

The sequence is 5, 7, 11, 19 because each term is doubled and then reduced by 3.

105. TACHS-M1-005 - Answer A: $3n - 7$

"Three times n" is $3n$, and "seven less than" means subtract 7.

106. TACHS-M1-006 - Answer B: 6 hours

Subtract the \$6 initial fee, then divide: $(\$30 - \$6) \div \$4 = 6$ hours.

107. TACHS-M1-007 - Answer C: 85

The total must be $5 \times 84 = 420$; the known scores total 335, so the missing score is 85.

108. TACHS-M1-008 - Answer D: $1/5$

The probability is $3/6 \times 2/5 = 6/30 = 1/5$.

109. TACHS-M1-009 - Answer A: 65°

Triangle angles total 180° , so the third angle is $180^\circ - 47^\circ - 68^\circ = 65^\circ$.

110. TACHS-M1-010 - Answer B: 120 cm^2

Since $2(15 + \text{width}) = 46$, the width is 8; therefore, area = $15 \times 8 = 120 \text{ cm}^2$.

111. TACHS-M1-011 - Answer C: 8

Convert 2.4 meters to 240 centimeters, then calculate $240 \div 30 = 8$ pieces.

112. TACHS-M1-012 - Answer D: 8:45 p.m.

Adding 1 hour gives 7:47 p.m.; adding 58 minutes gives 8:45 p.m.

113. TACHS-M1-013 - Answer D: $3 \frac{3}{4}$ cups

$2 \frac{1}{4} \div 18 = 1/8$ cup per bar, and $30 \times 1/8 = 3 \frac{3}{4}$ cups.

114. TACHS-M1-014 - Answer A: 42

The common factors of 84 and 126 include 42, and no larger number divides both.

115. TACHS-M1-015 - Answer B: -3°C

Starting at -6 , the changes give $-6 + 11 - 8 = -3$.

116. TACHS-M1-016 - Answer C: (3, 5)

Substituting $x = 3$ gives $y = 3(3) - 4 = 5$.

117. TACHS-M1-017 - Answer D: $x = 9$

Expanding gives $5x - 10 = 3x + 8$, so $2x = 18$ and $x = 9$.

118. TACHS-M1-018 - Answer B: The mean increases by 5, and the range stays the same.

Adding 5 to every value adds 5 to the mean but does not change the difference between extremes.

119. TACHS-M1-019 - Answer A: $3/8$

The favorable outcomes are 4, 6, and 8, giving 3 favorable sections out of 8.

120. TACHS-M1-020 - Answer D: 30%

The total is 180 books, and $54 \div 180 = 0.30 = 30\%$.

121. TACHS-M1-021 - Answer C: 15 cm

The hypotenuse is $\sqrt{(9^2 + 12^2)} = \sqrt{225} = 15$ centimeters.

122. TACHS-M1-022 - Answer A: 44 inches

Circumference equals πd , so $22/7 \times 14 = 44$ inches.

123. TACHS-M1-023 - Answer B: 324 square units

The first side is 9; doubling it gives 18, and $18^2 = 324$.

124. TACHS-M1-024 - Answer C: 63 miles

The actual distance is $3 \frac{1}{2} \times 18 = 63$ miles.

125. TACHS-M1-025 - Answer D: 240
The box's volume is $10 \times 6 \times 4 = 240$ cubic inches, so it holds 240 unit cubes.
126. TACHS-M1-026 - Answer A: $\frac{2}{3}$
Dividing by $\frac{9}{10}$ means multiplying by $\frac{10}{9}$: $\frac{3}{5} \times \frac{10}{9} = \frac{2}{3}$.
127. TACHS-M1-027 - Answer C: \$39
A 25% reduction leaves \$36, and delivery raises the total to \$39.
128. TACHS-M1-028 - Answer B: 9
The positive factors of 9 are 1, 3, and 9.
129. TACHS-M1-029 - Answer D: 11
Expanding gives $4x - 12 = 2x + 10$, so $2x = 22$ and $x = 11$.
130. TACHS-M1-030 - Answer A: 34
Substitute position 12 into the rule: $3 \times 12 - 2 = 34$.
131. TACHS-M1-031 - Answer D: $4 + 2.50m \leq 24$
The fixed fee plus mileage cost must be less than or equal to \$24.
132. TACHS-M1-032 - Answer D: 10
A mean of 6 requires a total of 30; the known values total 20, so $n = 10$.
133. TACHS-M1-033 - Answer A: $\frac{1}{15}$
The probability is $\frac{2}{6} \times \frac{1}{5} = \frac{2}{30} = \frac{1}{15}$.
134. TACHS-M1-034 - Answer B: 24 square units
The area is $\frac{1}{2} \times 6 \times 8 = 24$ square units.
135. TACHS-M1-035 - Answer B: 8
The exterior angles total 360° , so the polygon has $360 \div 45 = 8$ sides.
136. TACHS-M1-036 - Answer C: 10
Convert 2.5 liters to 2,500 milliliters, then compute $2,500 \div 250 = 10$.
137. TACHS-M1-037 - Answer C: 63 miles
Multiply the map distance by the scale: $3.5 \times 18 = 63$ miles.
138. TACHS-M1-038 - Answer C: 80 liters
The added 24 liters equal $\frac{9}{10} - \frac{3}{5} = \frac{3}{10}$ of capacity, so capacity is $24 \div \frac{3}{10} = 80$.
139. TACHS-M1-039 - Answer B: 40
Multiplying both numbers by 100 gives $240 \div 6 = 40$.
140. TACHS-M1-040 - Answer D: \$54
A 20% reduction makes the price \$60, and the \$6 coupon lowers it to \$54.
141. TACHS-M1-041 - Answer A: 7
Solve $3x - 4 = 17$: $3x = 21$, so $x = 7$.
142. TACHS-M1-042 - Answer D: 35
The increases are 3, 6, and 9, so the next increase is 12 and $23 + 12 = 35$.
143. TACHS-M1-043 - Answer C: 85
Five scores total $5 \times 84 = 420$; the four known scores total 335, leaving 85.
144. TACHS-M1-044 - Answer C: $\frac{1}{22}$
The probability is $\frac{3}{12} \times \frac{2}{11} = \frac{6}{132} = \frac{1}{22}$.
145. TACHS-M1-045 - Answer A: 28
Six trips must total $6 \times 23 = 138$; the first five total 110, so the sixth had 28.
146. TACHS-M1-046 - Answer B: 115°
An exterior angle equals the sum of the two remote interior angles: $48^\circ + 67^\circ = 115^\circ$.
147. TACHS-M1-047 - Answer C: 44 centimeters
Circumference equals πd , so $\frac{22}{7} \times 14 = 44$ centimeters.
148. TACHS-M1-048 - Answer A: 15 units
Since $9^2 + 12^2 = 81 + 144 = 225$, the hypotenuse is $\sqrt{225} = 15$.
149. TACHS-M1-049 - Answer B: 2.1 liters
Six batches use $6 \times 350 = 2,100$ milliliters, which equals 2.1 liters.

150. TACHS-M1-050 - Answer D: 8:43 p.m.

Adding 1 hour gives 7:47 p.m.; adding 56 minutes gives 8:43 p.m.

Ability

151. TACHS-A1-001 - Answer A: ■■■

Moving right fills the shape and adds one; moving down changes circles to squares without changing count.

152. TACHS-A1-002 - Answer B: → □

The arrow rotates clockwise across the row, while the shape changes from circle to square down the column.

153. TACHS-A1-003 - Answer C: ☆★

Each third cell places the first cell's figure before the second cell's figure.

154. TACHS-A1-004 - Answer D: Top: □; Bottom: ■

The positions exchange across the row, and circles become squares down the columns while fill is preserved.

155. TACHS-A1-005 - Answer A: △△△△△

Counts increase by one from left to right, so four triangles must become five.

156. TACHS-A1-006 - Answer B: Two holes near the upper-left and upper-right corners

Unfolding reflects the punched hole horizontally across the vertical centerline.

157. TACHS-A1-007 - Answer C: Four holes placed symmetrically across both centerlines

Two perpendicular half-folds create four layers, so unfolding reflects the hole across both centerlines.

158. TACHS-A1-008 - Answer D: One diamond-shaped opening centered on the vertical centerline

The cut on the folded edge reflects across the fold, joining two triangular halves into a centered diamond.

159. TACHS-A1-009 - Answer A: Two holes mirrored across the upper-left-to-lower-right diagonal

A single diagonal fold produces one reflected partner across that same diagonal.

160. TACHS-A1-010 - Answer B: ◇ ◆ ◇

Each group has matching open outer shapes and the filled form of that shape in the center.

161. TACHS-A1-011 - Answer C: ← ↑ →

In every valid arrangement, each arrow is obtained by rotating the previous arrow 90° clockwise.

162. TACHS-A1-012 - Answer D: ◇ inside ☆

Each valid figure places one open shape inside a different open shape.

163. TACHS-A1-013 - Answer B: | with □ on the right

The shape crosses to the opposite side, while moving down changes the shape and rotates the line.

164. TACHS-A1-014 - Answer B: □

A square completes the set of three different shapes in both the final row and final column.

165. TACHS-A1-015 - Answer A: ●△

The first shape becomes filled across the row, and the second changes from square to triangle down the column.

166. TACHS-A1-016 - Answer C: ↑

Continuing the 90° clockwise rotation after the left arrow produces an up arrow.

167. TACHS-A1-017 - Answer A: Two holes near the upper-left and lower-left corners

Unfolding across the horizontal centerline reflects the punched hole vertically on the left side.

168. TACHS-A1-018 - Answer C: Two holes mirrored across the upper-right-to-lower-left diagonal

Reversing the diagonal fold reflects the hole once across that same diagonal.

169. TACHS-A1-019 - Answer D: Four holes, one in each quadrant, symmetric across both centerlines

Two perpendicular folds create four reflected locations, one in each quadrant.

170. TACHS-A1-020 - Answer B: Two matching semicircular notches, one in each half of the top edge

Unfolding reflects the outer-edge cut to the corresponding position on the other half.

171. TACHS-A1-021 - Answer D: + × + ×

Each valid arrangement alternates two distinct open symbols in an ABAB pattern.

172. TACHS-A1-022 - Answer A: Top: □; Center: ○; Bottom: □

The top and bottom symbols match, while the center symbol is different.

173. TACHS-A1-023 - Answer C: ◇ inside ☆; ◆ outside ☆

Each group places an open shape inside a boundary and its filled counterpart outside.

174. TACHS-A1-024 - Answer D: ← ← ↑

The first direction repeats, and the third arrow is its 90° clockwise rotation.

175. TACHS-A1-025 - Answer C: ◆
Only ◆ appears in both of the first two cells in row 3.
176. TACHS-A1-026 - Answer A: △↑
The arrow rotates clockwise from ← to ↑, and the filled triangle returns to open.
177. TACHS-A1-027 - Answer B: △ △ △ △ △
The third cell combines two and three triangles, producing five open triangles.
178. TACHS-A1-028 - Answer D: ● ★
Swapping ☆ ○ gives ○ ☆, then filling both shapes gives ● ★.
179. TACHS-A1-029 - Answer C: →
The middle ↑ is opposite the top ↓; rotating ↑ clockwise gives →.
180. TACHS-A1-030 - Answer B: At all four corners
Reversing both centerline folds reflects the corner punch into all four corners.
181. TACHS-A1-031 - Answer A: Two holes on the vertical centerline, equally spaced above and below the horizontal centerline
Unfolding the horizontal fold reflects the punch vertically while preserving its centerline position.
182. TACHS-A1-032 - Answer B: There will be two holes reflected across the diagonal.
A punch away from a single diagonal fold creates a mirror pair across that diagonal.
183. TACHS-A1-033 - Answer C: 4
The corner at both folded centerlines reflects across each fold, creating four cuts around the center.
184. TACHS-A1-034 - Answer A: ◇◆◇
Each group has matching open outer shapes and the filled version of that shape in the center.
185. TACHS-A1-035 - Answer B: Top: ◆; Bottom-left: △; Bottom-right: △
The two bottom shapes match each other and differ from the single top shape.
186. TACHS-A1-036 - Answer D: ← ↑ →
Each arrow is obtained by rotating the previous arrow 90° clockwise.
187. TACHS-A1-037 - Answer D: ☆+☆
Each figure has identical open shapes on both sides of one central line symbol.
188. TACHS-A1-038 - Answer A: Top-left: △
The triangle advances clockwise from bottom-right to bottom-left and then to top-left.
189. TACHS-A1-039 - Answer C: ☆ inside ○
The third cell places the first shape inside the second, so the star goes inside the circle.
190. TACHS-A1-040 - Answer D: ☆ ○ ☆
Only the center changes open to filled to open, while the two outer stars remain unchanged.
191. TACHS-A1-041 - Answer B: △|||
Each step downward adds one line, so the final cell has four lines after the triangle.
192. TACHS-A1-042 - Answer A: Two holes in the upper half, reflected across the vertical centerline
Reversing the single vertical fold reflects the punch horizontally while preserving its upper-half position.
193. TACHS-A1-043 - Answer D: 2
The punch lies on the vertical fold, so that fold creates no duplicate; reversing the horizontal fold creates two holes.
194. TACHS-A1-044 - Answer B: One complete circular hole centered on the diagonal
A semicircular cut on the folded diagonal reflects into its matching half when the diagonal fold is opened.
195. TACHS-A1-045 - Answer D: At all four original corners
The punched original corner reflects across both center folds, producing one matching hole at every original corner.
196. TACHS-A1-046 - Answer C: ◇|◆
Each group places an open shape left of a divider and its filled counterpart on the right.
197. TACHS-A1-047 - Answer B: Outer: ☆; Inner: ◇; symbol below: ◇
The symbol below must exactly match the inner shape while differing from the outer shape.
198. TACHS-A1-048 - Answer A: ←|↑|→|↓
Each arrow turns 90° clockwise from the preceding arrow; choice A continues that four-direction cycle.
199. TACHS-A1-049 - Answer D: ☆/☆
The first and third symbols are identical shapes, with a different line symbol between them.

200. TACHS-A1-050 - Answer C: Top: ○; Center: ☆; Bottom: ○

The top and bottom shapes must match, while the center must be a different open shape.